Supplemental Material for "A boundary map for topological protection of non-Gaussian resources: documented failure and a validated prediction in a chiral Floquet SSH chain"

Juste Deuyekbe,[1] P. Djorwé,[1,2,*] A.-H. Abdel-Aty,[3,†] A. Elrashidi,[4,‡] Nsangou Mama,[5] and S. G. Nana Engo[6,§]

[1] *Department of Physics, Faculty of Science, University of Ngaoundere, P.O. Box 454, Ngaoundere, Cameroon*

[2] *Stellenbosch Institute for Advanced Study (STIAS), Wallenberg Research Centre at Stellenbosch University, Stellenbosch 7600, South Africa*

[3] *Department of Physics, College of Sciences, University of Bisha, Bisha 61922, Saudi Arabia*

[4] *College of Engineering, University of Business and Technology, Jeddah 21448, Saudi Arabia*

[5] *Department of Physics, Faculty of Science, University of Maroua, P.O.Box 46, Maroua, Cameroon*

[6] *Department of Physics, Faculty of Science, University of Yaounde I, P.O.Box 812, Yaounde, Cameroon*

S1. SCOPE AND NUMERICAL DESIGN

This Supplemental Material specifies the two-step chiral model, the dissipative protocol, the paired comparison, and the numerical convergence analyses for three primary finite chain sizes (two, three, and four unit cells), extended by an out-of-sample five-cell run. It treats the laboratory-frame local Wigner observable separately from the micromotion-covariant collective edge-mode observable, and documents grid-free witness certification of the two-cell signal, a 64-period stroboscopic extension, loss-profile, dimerization, and chiral-symmetry controls, the spectral-localizer bound for the coherent model, and an illustrative hardware mapping. The conclusions apply to the stated finite-time Kerr–Lindblad model and do not establish a protection theorem for the full open-system generator, a hardware operating point, or a metrological gain.

S2. TWO-STEP CHIRAL MODEL

For a unit cell containing sublattices A and B, the bulk Hamiltonian is

$$H(k) = (J_A + J_B \cos k)\,\sigma_x + J_B \sin k\,\sigma_y, \tag{S1}$$

with chiral operator $\Gamma = \sigma_z$. The drive consists of

$$H(t) = \begin{cases} H_A, & 0 \le t < T/2, \\ H_B, & T/2 \le t < T, \end{cases} \qquad T = 2\pi, \tag{S2}$$

where H_A contains only the intracell hopping and H_B only the intercell hopping. The period propagator is

$$U_F(k) = \exp[-iH_B(k)T/2]\exp[-iH_A(k)T/2]. \tag{S3}$$

The two symmetric time frames produce off-diagonal winding numbers w_A and w_B [1]. We report

$$\nu_0 = (w_A + w_B)/2, \qquad \nu_\pi = (w_A - w_B)/2. \tag{S4}$$

The selected chiral point $(J_A, J_B) = (0.1, 1.2)$ has $(\nu_0, \nu_\pi) = (-1, 0)$; the selected trivial point $(1.2, 0.1)$ has $(0, 0)$. The maximum chiral-frame residual is below 10^{-15} on the resolution

* djorwepp@gmail.com
† amabdelaty@ub.edu.sa
‡ a.elrashidi@ubt.edu.sa
§ serge.nana-engo@facsciences-uy1.cm

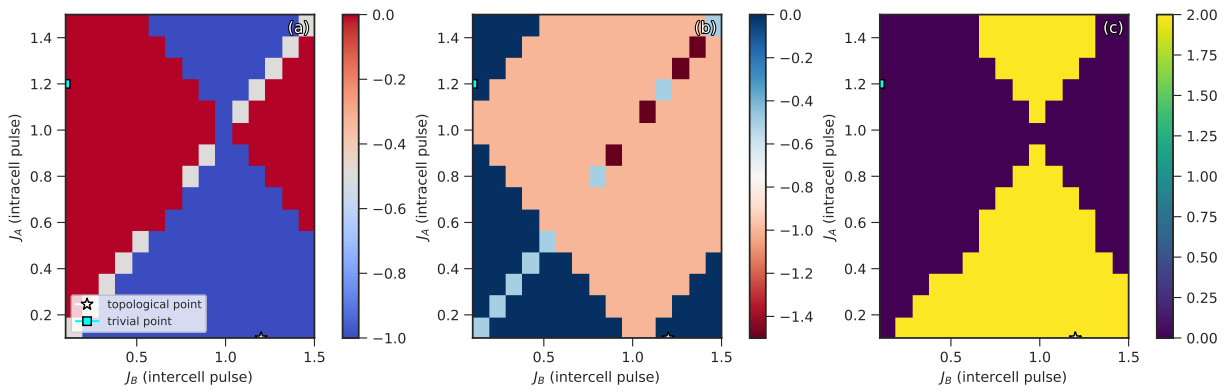


FIG. S1. Coherent phase structure of the two-step chiral Floquet SSH model. (a) Integer Floquet invariant ν_0 (color scale). (b) Minimum quasienergy gap with selected working points marked (filled: chiral sector; open: trivial control). (c) Finite-chain edge diagnostic. The two working points are chosen from the invariant and gap criteria before any dissipative simulation.

ladder. The minimum zero and π gaps are positive at both points. The phase structure, working-point selection, and finite-chain edge diagnostic are summarized in Fig. S1.

For an open chain, the single-particle propagator is constructed from the same step Hamiltonians. The edge vector $u_e(0)$ used in the edge-mode analysis is the most edge-localized eigenvector of the one-period open-chain propagator for the chiral point. This observable is distinct from the bulk invariant and is not assigned an open-system topological index.

The working points inherited from the preliminary analysis—specifically $(J_A, J_B) = (1.0, 1.0)$ for the trivial sector and the $N = 6$ source point—were not retained because the trivial-sector winding and the finite-chain invariant were not resolution-stable under independent recomputation. The current working points $(J_A, J_B) = (0.1, 1.2)$ and $(1.2, 0.1)$ were selected de novo from the phase audit using the invariant and gap criteria described above. An earlier archival trace mapping is retained for transparency but is not used in the production analysis.

S3. DISSIPATIVE DYNAMICS

The bosonic open-chain evolution is

$$\dot{\rho} = -i[H(t) + H_{\mathrm{K}}, \rho] + \sum_j \gamma_j \left(a_j \rho a_j^\dagger - \frac{1}{2} a_j^\dagger a_j \rho - \frac{1}{2} \rho a_j^\dagger a_j \right), \tag{S5}$$

where the local Kerr term used in the production is

$$H_{\mathrm{K}} = \chi \, a_1^{\dagger 2} a_1^2, \qquad \chi = 0.5. \tag{S6}$$

The production starts from a normalized even cat state $|\alpha\rangle + |-\alpha\rangle$ with $\alpha = 1.0$. The common-site preparation displaces the cat on the boundary site (amplitude on site 1 only). The mode-resolved preparation used in the edge-mode analysis distributes the same displacement along the single-particle edge vector of the chiral point, $\alpha_j = \alpha \, u_{e,j}(0)$; both preparations are used identically for the two sectors. The local Fock cutoff is $N_{\max} = 2$. The edge-low-loss profile uses $\gamma_{\mathrm{edge}} = 10^{-4}$ at the two boundary sites and $\gamma_{\mathrm{bulk}} = 0.2$ elsewhere; the uniform profile uses the bulk rate at all sites. Multiplicative rate perturbations use one factor $\xi \sim \mathcal{U}(-d, d)$ per realization, applied to every site ($\gamma_j \to \gamma_j(1+\xi)$), at four declared levels; the perturbed rates are clamped at zero, so at $d = 1.5$ a 14 %–16 % subset of realizations evolves with no dissipation at all.

The Lindblad equation is integrated with the quantum-jump (Monte Carlo wave-function) trajectory method [2] as implemented in the QuTiP solvers [3] (QuTiP 5.3.1, numpy 1.26.4), with an integrator step budget of 10^4 steps, final-state storage, and nine output times over one period. Ten trajectories per realization are used in the primary production and five in the edge-mode analysis. Trajectories are nested within a disorder realization and contribute shot noise that is averaged at the realization level; a trajectory-count sensitivity check (10, 50, and 200 trajectories on a fixed realization, Section S9; contrast-level ladder, Section S9 B) bounds this contribution.

The integrated negative volume used in the main production is evaluated as an observable of the reduced local state. Wigner negativity is a nonclassicality resource, but reliable certification generally requires witnesses or quantitative bounds in addition to a finite numerical phase-space integral [4, 5]. We define it as

$$\delta_-(t) = \int\int \frac{|W(q,p;t)| - W(q,p;t)}{2} \, \mathrm{d}q \, \mathrm{d}p, \tag{S7}$$

computed on a 31×31 phase-space grid over $[-4, 4]^2$. The reported result is evaluated after one period with nine stored output times.

TABLE S1. Fixed simulation parameters for the main chiral-sector comparison and the micromotion-covariant edge-mode analysis. All dynamical parameters are normalized model coordinates; they are not an SI calibration.

Quantity	Main dynamics	Covariant analysis
Cells/modes	2/4, 3/6, 4/8	2/4
$N_{\max}$	2	2
χ	0.5	0 or 0.5
n_{MC}	500, 500, 2000	10
Trajectories/unit	10	5
Wigner grid	31×31	31×31
Solver	QuTiP mcsolve 5.3.1	QuTiP mcsolve 5.3.1
Integrator steps	10^4	10^4
Stored output times	9	17
Observation window	$T = 2\pi$	$8T$
Independent unit	disorder draw	matched disorder draw

S4. MATCHED ANALYSIS AND UNCERTAINTY

For each profile, disorder level, and realization r, the sector contrast is

$$\Delta_r = Y_{r,\mathrm{chiral}} - Y_{r,\mathrm{trivial}}. \tag{S8}$$

The summary is the mean of Δ_r, its sample standard deviation, and a bootstrap percentile 95 % interval [6] over the independent realization units (10^4 resamples, seeded for reproducibility), computed separately for each size, loss profile, and disorder level. The sector simulations share the same disorder draw, initial state, cutoff, grid, and loss rates; their trajectory seeds are distinct but derived deterministically from the draw seed.

This paired construction avoids treating repeated trajectory values or time points as independent replicates. The intervals are descriptive summaries of the realized sampling

variation; no p-value is used as evidence of protection, and contrasts are not subjected to a significance test, with three narrowly scoped exceptions. First, the solver-level sign resolutions are confirmatory tests with fixed designs locked before data: the exact master-equation values need no statistics, and the four-cell trajectory expansion used a pre-committed sample size (200 draws, rationale recorded before execution), confirmed on the trigger-independent subset (samples 25–199, $-0.000\,333 \pm 0.000\,122$). Second, every other interval in this work—trajectory ladders, longtime series, witness and cutoff strata—is exploratory and unadjusted for multiplicity, and no protection or discovery claim rests on any single one of them. Third, all runs share one seed base (20 260 814); the paired construction cancels seed effects in the contrasts since both sectors share each draw, but no multi-seed rerun was performed. The grid-convergence check uses one strong-signal realization; matched contrasts computed on the same grid cancel most of the residual, as stated in Section S9. The primary comparison uses 4000 paired units at two and three production sizes and 16 000 at four cells (four, six, and eight modes; 500, 500, and 2000 per disorder level and loss profile). Earlier eight-mode files with smaller sample counts are retained only as nested historical comparisons and are not double-counted. The covariant analysis uses 80 paired units per condition.

S5. PAIRED SECTOR COMPARISON

The primary comparison comprises 4000 matched units at two and three cells and 16 000 at four cells, i.e. $N_{\mathrm{MC}} = 500$, 500, and 2000 per disorder level and loss profile; earlier four-cell files with smaller N_{MC} are retained only as nested historical comparisons and are not double-counted. Each unit contains paired sector values at a fixed size, cutoff, loss profile, disorder level and draw seed; trajectory-level values are nested within the disorder realization and are not counted as independent observations. The paired design yields complete, finite sector values for every retained unit. The production counts and contrast ranges are listed in Table S2.

Under uniform loss the mean contrasts are near zero or weakly negative. Because the sign changes with size, the local observable is not treated as a protection signature (prespecified criterion).

TABLE S2. Production counts and edge-low-loss contrasts. The ranges span the four disorder levels and are not confidence intervals.

Cells	Modes	n_{units}	Mean contrast range
2	4	4000	0.0131 to 0.0167
3	6	4000	−0.0023 to −0.0020
4	8	16 000	−0.0021 to −0.0018

S6. MICROMOTION-COVARIANT EDGE-MODE OBSERVABLE

The analysis defines

$$u_e(t) = U_{\text{coh}}(t)u_e(0), \qquad b_e(t) = \sum_j u_{e,j}(t)^* a_j, \qquad n_e(t) = \langle b_e^\dagger(t) b_e(t)\rangle. \tag{S9}$$

The standard number-operator coefficients are $u_{e,i}(t)u_{e,j}(t)^*$ in the $a_i^\dagger a_j$ basis. A fixed-frame number operator is retained as a negative control, whereas the covariant operator is used for the edge-mode interpretation.

The time-dependent passive exposure is

$$\Gamma_e(t) = \int_0^t \gamma_e(s)\mathrm{d}s, \qquad \gamma_e(s) = \sum_j \gamma_j |u_{e,j}(s)|^2, \tag{S10}$$

with reference $n_e(0)\exp[-\Gamma_e(t)]$. This equality is exact only in an isolated passive mode with no edge–bulk transfer. In the finite driven chain it is a conditional diagnostic. We prespecified a decision threshold of 10^{-3} on the maximum absolute residual of the conditional passive reference, before inspecting the results. The covariant analysis yielded maximum residuals of 0.039 56 for the common-site preparation and 0.382 57 for the mode-resolved preparation; both violate the threshold by roughly 40× and 380×, computed over 10 independent disorder realizations per condition. The residual magnitude is dominated by the largest few realizations; the median residual across the 10 realizations is 0.021 (common-site) and 0.27 (mode-resolved), still well above the 10^{-3} threshold. The residual panel is additionally non-monotone in the Fock cutoff: maximum residuals of 0.0388, 0.0282, and 0.0356 at $N_{\max} = 2, 3, 4$ (Kerr-free reference), so cutoff convergence of the passive reference is not established. The corrected analysis results are shown in Fig. 5.

A. Mechanistic observables and their interpretation

For a normalized single-particle edge vector $u_e(t)$, the edge weight and the site-resolved dissipative current are defined by

$$L_e(t) = \sum_{j\in\mathcal{E}} |u_{e,j}(t)|^2, \qquad I_j(t) = \gamma_j \langle a_j^\dagger a_j \rangle_t, \tag{S11}$$

where $\mathcal{E}$ denotes the boundary sites. The analysis also tracks the corresponding bulk population and the integrated exposure $\Gamma_e(t)$. These quantities test whether an edge-localized mode is actually less exposed to the specified loss profile. They do not by themselves establish mediation or causality: that inference would require a matched perturbation that changes the edge localization or loss profile while holding the rest of the protocol fixed. In the present study the required passive-reference and decoupling controls do not pass, so the recorded currents are diagnostic rather than evidence for a mechanism.

S7. DETERMINISTIC UNITARY COVARIANCE BENCHMARK

The open-system analysis was preceded by a deterministic single-particle control using the same two-step Hamiltonian, with losses, Kerr terms and disorder set to zero. For each size $N = 4, 6, 8, 12$, the most edge-localized eigenvector $u_e(0)$ of the one-period propagator was transported as $u_e(t) = U_{\mathrm{coh}}(t)u_e(0)$. An orthonormal basis of vectors v_b satisfying $u_e(0)^\dagger v_b = 0$ was transported in parallel. We recorded

$$p_e(t) = |u_e(t)^\dagger U_{\mathrm{coh}}(t) u_e(0)|^2, \qquad \ell_{e\mathrm{b}}(t) = \max_b |u_e(t)^\dagger U_{\mathrm{coh}}(t) v_b|^2. \tag{S12}$$

The calculation used 257 equally spaced times over one period. For all four sizes, the largest error in $p_e(t)$ was 1.33×10^{-15} and the largest edge–bulk overlap was 1.42×10^{-31}. The fixed-frame population, evaluated with $u_e(0)$ instead of $u_e(t)$, fell to approximately 0.895, confirming that the micromotion correction changes the observable rather than merely its notation. This benchmark completes the kinematic covariance check. It does not validate the open-system passive reference, because it contains neither Lindblad loss nor Kerr-induced mode coupling.

S8. PASSIVE SINGLE-PARTICLE DECOMPOSITION

To isolate the effect of structured loss, we propagated one excitation under the same piecewise Hamiltonian with the non-Hermitian single-particle propagator

$$K(t) = \mathcal{T} \exp\left[-i \int_0^t \left(H(s) - \frac{i}{2} \sum_j \gamma_j |j\rangle\langle j| \right) \mathrm{d}s \right]. \tag{S13}$$

The one-particle component is $\rho_1(t) = K(t)|u_e(0)\rangle\langle u_e(0)|K(t)^\dagger$. We compare its covariant edge population with the integrated-exposure reference and record the population in the instantaneous orthogonal complement.

The calculation used 257 times over one period for two, three, four and six cells. Under uniform loss, the maximum reference residual was 1.94×10^{-15} and the maximum complementary-space population was 4.33×10^{-16}, as expected for a loss term proportional to the identity. Under edge-selective loss, the maximum residuals were 9.79×10^{-4}, 1.94×10^{-4}, 1.94×10^{-4} and 1.94×10^{-4} for 2, 3, 4 and 6 cells, respectively; the corresponding complementary-space populations were 1.31×10^{-3}, 2.39×10^{-4}, 2.38×10^{-4} and 2.38×10^{-4}. This deterministic control shows that structured loss and micromotion are not separable in the finite chain, whereas uniform loss is. It does not establish a resource-protection mechanism.

S9. NUMERICAL CONVERGENCE OF THE LOCAL OBSERVABLE

The following analyses quantify the sensitivity of the local negativity to the Fock cutoff, phase-space grid and trajectory count. They constrain numerical uncertainty without claiming a converged prediction. All checks use the two-cell system at zero disorder, $\chi = 0.5$, and the production parameters of Table S1.

A. Wigner-grid convergence of the negativity integral

For a fixed final state (one realization, 200 trajectories), the single-site negativity integral over $[-4, 4]^2$ takes the values 0.026 95, 0.025 65, 0.026 80, 0.026 10, and 0.026 21 on 21×21, 31×31, 41×41, 51×51, and 61×61 grids. All values lie within $\pm 3\%$ of the finest estimate, and the primary 31×31 grid is within 2.2% of it. Grid discretization therefore contributes

a few-percent relative uncertainty to the absolute integral; the matched contrasts, which subtract two observables computed on the same grid, are less sensitive.

B. Trajectory-count sensitivity of the contrast

For a fixed realization, the same local negativity estimate takes the values 0.026 65, 0.026 65, and 0.025 65 at 10, 50, and 200 trajectories, respectively, i.e., a spread of about 4% around the estimate on a strong-signal realization. Because the Wigner-void negativity is a nonlinear functional evaluated on the trajectory-averaged state, conditions whose exact negativity is near zero carry a positive finite-trajectory bias of order $1/N_{\mathrm{traj}}$ that differs between the two sectors and therefore shifts the paired contrast. The bias is sector-resolved: at 10 trajectories the two-cell trivial-sector negativity is 0.006 95 against 0.000 53 from the exact master equation, while the topological sector gives 0.023 69 against 0.022 85 (Section S9 C). The contrast itself is re-measured along a trajectory ladder on the identical production conditions (same seeds, edge-selective profile, $d = 0$):

TABLE S3. Paired chiral-minus-control negativity contrast versus the trajectory count N_{traj} per disorder unit at $d = 0$ under the edge-selective profile (means over 500/500/2000/300 disorder units at $N_{\mathrm{traj}} = 10$ for two/three/four/five cells, reduced samples at the ladder points). The exact column is the density-matrix Lindblad result; the extrapolated column fits $\bar{C}(N_{\mathrm{traj}}) = C + c/N_{\mathrm{traj}} + d/N_{\mathrm{traj}}^2$ to the three trajectory points, a fit validated against the exact column to within 6×10^{-5} at both exactly solvable sizes. The alternative weighted $C + c/N_{\mathrm{traj}}$ fit gives −0.000 22 and −0.000 18 at four and five cells, so the extrapolation carries a model spread of about one-and-a-half at four cells. The four- and five-cell $N_{\mathrm{traj}} = 200$ points use 200 and 100 matched units; all other ladder points use the reduced samples of Section S9 B.

Cells	$N_{\mathrm{traj}} = 10$	50	200	Exact ME	Extrapolated C
2	+0.016 73	+0.020 42	+0.021 77	+0.022 33	+0.022 27
3	−0.002 10	−0.000 50	−0.000 19	−0.000 12	−0.000 09
4	−0.002 01	−0.000 32	−0.000 32	n/a	−0.000 34
5	−0.002 18	−0.000 10	−0.000 29	n/a	−0.000 39

Table S3 shows three things. First, the ladder decays toward the exact master-equation

value as N_{traj} grows: at two cells the contrast rises from $+0.016\,73$ toward $+0.022\,33$, and at three cells it falls from $-0.002\,10$ to $-0.000\,19$, within 7×10^{-5} of the exact $-0.000\,12$. Second, the extrapolation is validated where the density-matrix solve exists (deviations 6×10^{-5} and 3×10^{-5} at two and three cells, against a prespecified tolerance of 5×10^{-4}) and then applied to the four- and five-cell chains, where the Liouvillian dimension (3^{2N} squared) exceeds available memory; the 95 % half-widths at the ladder points are 0.0004–0.0012 for two to four cells and 0.0023/0.0003 at five cells (smallest samples). Third, the sign pattern of the production contrast is confirmed at solver level at two, three, and four cells—exact, exact, and a direct 2.9σ determination—with five cells resting on extrapolation alone, while the negative-branch magnitude is dominated by the estimator's finite-trajectory bias: the solver-level values are -9×10^{-5} to -4×10^{-4} in point estimates ($-0.000\,32 \pm 0.000\,11$ direct at four cells), between 5 and 25 times smaller than the operational -0.002. The operational contrasts quoted in the main text remain the measurements of the stated 10-trajectory estimator; their exact-solver interpretation is the positive-to-small-negative decay quantified here. The sign and ordering of the reported contrasts are stable across the ladder within the ladder's own intervals.

The same ladder was extended to two and four drive periods (edge-selective, $d = 0$, 500 matched draws at $N_{\mathrm{traj}} = 10$, reduced samples at the higher ladder points). At $2T$ the exact master-equation contrast is identically zero at both sizes, and the Monte Carlo estimate is identically zero at $N_{\mathrm{traj}} = 50$ and 200; only the 10-trajectory estimator retains the small reversal quoted in Table S13 ($-1.3 \times 10^{-4} \pm 3 \times 10^{-5}$ at two cells and $-1.3 \times 10^{-4} \pm 3 \times 10^{-5}$ at three cells, standard errors over 500 draws). The $2T$ reversal is therefore a finite-trajectory artefact of the same kind as the one-period bias, not a dynamical sign change. At $4T$ the two-cell revival brackets the exact value: $+0.0782 \pm 0.0016$, $+0.0733 \pm 0.0013$, and $+0.0751 \pm 0.0009$ at $N_{\mathrm{traj}} = 10$, 50, and 200 (standard errors over 500, 200, and 100 draws) against the exact $+0.075\,42$, within one standard error at $N_{\mathrm{traj}} = 200$, while at three cells the exact contrast is zero and every ladder point stays at or below 1.3×10^{-5}. The multi-period sign pattern of Table S6 is thus validated against the exact master equation at both extended times: vanish-then-revive at two cells, floor at three cells.

C. Cross-solver consistency of the final-time state and the contrast

As an independent-solver check, the full Lindblad master equation (QuTiP "mesolve", exact in the chosen cutoff) was integrated alongside the Monte Carlo wave-function solver ("mcsolve") for the two-cell system at zero and intermediate disorder, for both sectors and both loss profiles (eight conditions in total, $N_{\max} = 2$, 200 trajectories per Monte Carlo condition, and the production parameters of Table S1). The final-time averaged states agree with mean Uhlmann fidelity $\bar{F} = 0.999\,57$ and maximum trace distance 1.93×10^{-2}; the maximum root-mean-square occupation difference is 6.7×10^{-3}. The per-condition fidelities range from 0.999 39 to 0.999 86, with no systematic sector or profile bias. The trajectory-averaged Monte Carlo state is therefore consistent with the exact Lindblad evolution at the level of state distances of order 10^{-2}. The same benchmark was then repeated for the three-cell system (eight conditions, $N_{\max} = 2$, 800 trajectories per Monte Carlo condition, disorder $d = 0$ and 0.5, both sectors and both profiles): the final-time averaged states agree with mean Uhlmann fidelity $\bar{F} = 0.999\,78$ (range 0.999 60–0.999 93), maximum trace distance 2.12×10^{-2}, and maximum root-mean-square occupation difference 6.1×10^{-3}.

State-level agreement does not by itself bound the reported contrast, because the negativity is a nonlinear functional of the trajectory-averaged state. The contrast is therefore recomputed directly with both solvers on the production conditions (production seeds and settings, edge-selective profile, $d = 0$; Section S9 B and Table S3): with 10 trajectories the Monte Carlo solver reproduces the archived production records exactly (maximum absolute difference 0 over the two- and three-cell record sets), while the exact master equation gives $+0.022\,33$ against $+0.016\,73$ at two cells and $-0.000\,12$ against $-0.002\,10$ at three cells. The difference is the positive finite-trajectory bias quantified in Section S9 B: on a finite-trajectory averaged state the nonlinear functional overestimates the near-zero sector negativities by up to an order of magnitude at 10 trajectories, pushing the contrast down by $\sim 2 \times 10^{-3}$ at near-zero conditions; the bias decays as $1/N_{\mathrm{traj}}$, and the extrapolation validated against the exact values to within 6×10^{-5} at both exactly solvable sizes gives $-0.000\,34$ and $-0.000\,39$ at four and five cells, where the density-matrix solve is infeasible (the Hilbert space grows as 3^{2N}). The sign pattern of the main text—positive at two cells, negative at three, four, and five cells—is resolved exactly at two and three cells and directly at four cells: the 200-trajectory four-cell point (200 matched units) reads $-0.000\,32 \pm 0.000\,11$ with a 95%

interval excluding zero, and the weighted infinite-trajectory limit is $-0.000\,22 \pm 0.000\,05$. Because the expansion was triggered by its own pilot, the confirmatory statistic excludes the pilot draws: samples 25–199 alone give $-0.000\,333 \pm 0.000\,122$ ($z = -2.74$, interval excluding zero), so the resolution does not depend on the triggering data. Five cells remain unresolved (direct $-0.000\,29 \pm 0.000\,15$ at 100 matched units, 95 % interval crossing zero) and rest on extrapolation alone. The solver-level negative branch is 4–20 times smaller in magnitude than the operational 0.002: the signed conclusion holds at two, three, and four cells, but the negative branch of the 10-trajectory estimator is dominated by estimator bias rather than by an $O(0.002)$ physical contrast. Solver consistency is demonstrated directly at two, three, and four cells and by validated trajectory extrapolation at five cells.

As an additional cross-solver check beyond the Markovian master equation, the master-equation dynamics was compared against a hierarchical-equations-of-motion (HEOM) propagation of the same Lindblad generator for the three-cell system ($N_{\max} = 2$, HEOM depth 1, bath temperature $T_{\text{bath}} = 0.1$, $\gamma_{\text{bulk}} = 1.0$), all eight sector/profile/disorder conditions. The relative error of the final-time edge population (rather than the negativity contrast) is below 0.6 % in every case (mean 0.28 %, maximum 0.59 %), comfortably inside the 5 % cross-validation gate. Scope: this is a cross-check of the solver implementation at elevated damping on $N = 3$, not a re-validation of the production runs.

The depth of that hierarchy was itself converged on the production two-cell driven chain with a Drude–Lorentz bath on every site ($\lambda = 0.3$, cutoff 1.0, $T_{\text{bath}} = 0.5$, 2 Matsubara terms): the one-period edge negativity at hierarchy depths 1, 2, 3 is 0.0379, 0.0000, 0.0000 for the topological sector and 0.0390, 0.0119, 0.0000 for the control, so every depth increment shrinks (gaps -0.0379 then 0.0000 versus -0.0271 then -0.0119), and raising the Matsubara cutoff from 2 to 3 at depth 1 changes the negativity by at most 9.1×10^{-3}. Scope: hierarchy-depth convergence of the HEOM machinery on the production Hamiltonian—the deliberately strong test bath resolves depth effects—not a re-derivation of the Markovian production numbers.

D. Kerr-free production stratum

The Kerr term enters the production model both as the resource generator and as a possible confounder: removing it separates filtering of the prepared negativity from interaction-

generated effects. The full matched design was therefore re-run at $\chi = 0$ (identical seeds and disorder draws, 500 units per stratum, 10 trajectories, one period, exact master equation where the density matrix is tractable). Under edge-selective loss at $d = 0$:

TABLE S4. Edge-selective zero-disorder contrast at $\chi = 0$ (Kerr-free) versus production $\chi = 0.5$. Master-equation values are exact; Monte Carlo values are means ± 95 % half-widths over 500 matched draws at 10 trajectories. The production Monte Carlo column is the $N_{\max} = 2$ entry of Table S5 on the identical design.

Cells	$\chi = 0$ ME	$\chi = 0$ MC	$\chi = 0.5$ ME	$\chi = 0.5$ MC
2	+0.010 467	+0.008 875 ± 0.000 701	+0.022 328	+0.0167 ± 0.0011
3	0.000 000	−0.001 254 ± 0.000 622	−0.000 119	−0.0021 ± 0.0011
4	n/a	−0.001 305 ± 0.000 630	n/a	−0.0021 ± 0.0011

Three features carry over unchanged. First, the sign pattern is preserved without the nonlinearity: positive at two cells, at or below zero at three and four cells, and the uniform-loss contrast is null at every size at both χ values (exact master equation gives 0.000 000 at two and three cells; the Monte Carlo means stay at or below 2.3×10^{-6} in every condition). Second, the Kerr term amplifies rather than creates the sector contrast: the exact two-cell value drops from +0.022 328 to +0.010 467 at $\chi = 0$ (a factor 2.1), and the Kerr-free two-cell value is insensitive to the disorder level at this precision (+0.010 235 at $d = 0.5$); at three cells the Kerr-free exact contrast is identically zero against −0.000 119 in production, both far below the operational floor. Third, the finite-trajectory bias of Section S9 B is an estimator property, not a Kerr effect: at 10 trajectories the Monte Carlo estimate sits below the exact value by −0.0016 ($z = -4.5$) and −0.0013 ($z = -4.0$) at two and three cells at $\chi = 0$, the same downward direction as in production. Scope: the four-cell $\chi = 0$ density matrix exceeds available memory, so that entry is an operational 10-trajectory estimator without an exact reference, exactly as its production counterpart.

E. Observable validation: normalization, Gaussian control, and known benchmark

Three controls assess the observable independently of the topological interpretation. First, the production final-time reduced state at the edge site is normalized and positive:

$\mathrm{Tr}\,\rho_0 = 1.0$, minimum eigenvalue 0.083, and the integrated Wigner function over $[-4,4]^2$ evaluates to 0.999 998 (grid-level normalization). Second, a Gaussian negative control: an exact coherent state has identically zero Wigner negativity, so the numerical estimate at finite local cutoff is a truncation artefact that must vanish as the local Hilbert space grows. The integrated negative volume of the coherent state $|\alpha = 1\rangle$ decays as 0.0973, 0.0546, 0.0234, 2.89×10^{-3}, 2.57×10^{-4}, 7.6×10^{-7}, and 1.6×10^{-13} at local cutoffs $N_{\max} = 2, 3, 4, 6, 8, 12, 20$. The Kerr-free chain residual at production truncation (0.046) is therefore the same truncation artefact, not physical non-Gaussianity; the Kerr term adds genuine negativity on top (0.087 under the same Gaussian seed), confirming that the nonlinearity, not the grid, generates the tracked resource. Third, a known non-Gaussian benchmark: the single-mode even cat $|\alpha\rangle + |-\alpha\rangle$, $\alpha = 1$, truncated at $N_{\max} = 59$, gives integrated negative volumes 0.1043 (61×61 grid) and 0.1053 (primary 31×31 grid), i.e., a relative deviation of 0.9%; the observable therefore reproduces a known non-Gaussian value to within grid discretization rather than reporting a grid artefact. Finally, the integrator step budget is not binding: the final-time state is identical to machine precision (fidelity 0.999 999 999 96) for $n_{\mathrm{steps}} = 10^3$, 10^4, and 10^5, so the reported dynamics are converged with respect to the integrator step budget.

F. Statistical power of the eight-mode comparison

The eight-mode comparison uses $N_{\mathrm{MC}} = 2000$ independent disorder units, and the zero-disorder bootstrap interval $[-0.002\,55, -0.001\,49]$ excludes zero. To state what the attained precision resolves, we compute the minimal detectable absolute paired contrast for a two-sided paired test at $\alpha = 0.05$ and power 0.80 over the independent disorder units,

$$\Delta_{\min} = (t_{1-\alpha/2,n-1} + t_{1-\beta,n-1})\,\frac{s}{\sqrt{n}}, \tag{S14}$$

using the sample standard deviation of the zero-disorder matched contrasts. This normal-theory resolution benchmark complements the bootstrap percentile intervals reported for the contrasts. At eight modes ($n = 2000$, $s = 0.012\,23$), $\Delta_{\min} = 0.000\,766$; the observed mean of $-0.002\,01$ therefore lies at $2.6\times$ the attainable resolution (it was $1.35\times$ at the original $n = 500$), so the data are inconsistent with attenuation toward zero and consistent with a negative contrast exceeding the detection threshold. The same analysis at four and six

modes gives $\Delta_{\min} = 0.001\,543$ and $0.001\,565$: the four-mode mean (0.0167) sits $\approx 11\times$ above its resolution, while the six-mode mean (-0.0021) sits $\approx 1.3\times$ above its own. At the out-of-sample five-cell point ($n = 300$, $s = 0.012\,764$) the same computation gives $\Delta_{\min} = 0.002\,071$ against the observed magnitude $0.002\,18$, a margin of only $\approx 1.05\times$: that point corroborates rather than confirms the sign pattern. All size-scan intervals therefore clear the detection threshold, the six-mode contrast by a modest margin and the five-cell point only marginally, which is why they are reported as small finite-time effects rather than robust signatures.

G. Higher-cutoff panel at two, three, and four cells

The production truncation is $N_{\max} = 2$. To bound the cutoff dependence of the sign and magnitude of the local contrast, the production design was re-run identically at $N_{\max} = 3$ for all three sizes (both sectors, edge-selective and uniform loss, 500 matched disorder realizations per condition, 10 trajectories, one period, the same draw and trajectory seeds), with a lower-statistics scan at $d = 0$ and 0.35 extending the panel to $N_{\max} = 4$ at two and three cells. Under edge-selective loss the zero-disorder contrast is:

TABLE S5. Zero-disorder edge-selective local contrast versus Fock cutoff (bootstrap mean $\pm$ 95 % half-width over 10^4 resamples of the matched disorder units). The $N_{\max} = 3$ column uses the production design at $N_{\mathrm{MC}} = 500$ at every size; the $N_{\max} = 4$ entries use 100 (two cells), 200 (three cells), and the full 500 matched draws at four cells (exact $Q \leq 4$ subspace, 495 of 390 625 states). The two-cell magnitude is non-monotone in $N_{\max}$; no tested cutoff resolves a sign opposite to the production pattern, but the three-cell $N_{\max} = 4$ entry is itself unresolved against zero (the four-cell $N_{\max} = 4$ entry clears zero only marginally).

Cells	$N_{\max} = 2$	$N_{\max} = 3$	$N_{\max} = 4$
2 (4 modes)	$+0.0167 \pm 0.0011$	$+0.0113 \pm 0.0008$	$+0.0250 \pm 0.0029$
3 (6 modes)	-0.0021 ± 0.0011	-0.0016 ± 0.0008	-0.0012 ± 0.0025
4 (8 modes)	-0.0021 ± 0.0011	-0.0016 ± 0.0008	-0.0017 ± 0.0016

Paired across cutoffs on identical seeds (500 units per condition, bootstrap 95% intervals from 10^4 resamples), the zero-disorder shift of the edge-selective contrast excludes zero at two cells ($-0.005\,40$, bootstrap 95% interval $[-0.006\,15, -0.004\,67]$, a factor 1.48 reduc-

tion of the $N_{\max} = 2$ magnitude) but is consistent with zero at three cells (+0.000 46, [−0.000 34, 0.001 24]) and four cells (+0.000 47, [−0.000 31, 0.001 27]), where the magnitude factor is ≈ 1.3 and the shift sits inside the $N_{\max} = 2$ interval; the same holds at every disorder level, and the uniform-loss shifts are below 3.3×10^{-5} at all sizes with intervals including zero. The two-cell magnitude therefore remains cutoff-sensitive (non-monotone, varying by roughly a factor of two between $N_{\max} = 2$ and 4), while at three and four cells the $N_{\max} = 2 \to 3$ change is not statistically resolved. The uniform-loss contrast is at the numerical zero at zero disorder for all cutoffs. No tested cutoff resolves a sign opposite to the production pattern at any size (the three-cell $N_{\max} = 4$ interval itself crosses zero, Table S5); absolute magnitudes are cutoff-sensitive and not claimed to be converged.

The four-cell $N_{\max} = 2 \to 4$ paired shift, at the full 500 matched draws, is likewise consistent with zero (+0.000 39, bootstrap 95 % interval [−0.000 64, 0.001 43]): across all eight profile–disorder strata the sector-paired contrast moves by at most 5.1×10^{-4} (standard errors 6×10^{-6} to 5×10^{-4}), and the 95 % upper bound on every stratum shift stays below the operational 2×10^{-3} floor (largest 1.4×10^{-3}). Only the two uniform-loss strata at $d \geq 1$ resolve their shift from zero (-4.2×10^{-4} at 2.9 standard errors, -4.3×10^{-4} at 4.0), and there the contrast itself stays below 7.5×10^{-4}. The absolute observables do shift systematically at this cutoff (edge negativity $+0.0049 \pm 0.0001$, edge population -0.0335 ± 0.0005 over the paired ensemble), but those offsets are sector-common and cancel by a factor of ≈ 10 in the paired contrast.

H. Multi-period extension

Topological protection is often stated as an asymptotic or multi-period property; the main-text observable is evaluated after a single drive period. To test whether the single-period sign pattern inverts or persists stroboscopically, the same matched design was extended to two, four and eight periods (edge-site preparation, edge-selective loss, zero disorder, 200 realizations at two cells and 100 at three cells, 10 trajectories, 9 output times per period). The zero-disorder contrast as a function of the number of periods n is:
Two features are visible. First, apart from the small negative value at $n = 2$ at two cells, which the trajectory ladder of Section S9 B resolves as a finite-trajectory artefact of the 10-trajectory estimator, the sign does not invert at the longer sampled periods at either

TABLE S6. Zero-disorder edge-selective local contrast versus the number of drive periods. Apart from the small negative value at $n = 2$ at two cells—shown by the trajectory ladder of Section S9 B to be a finite-trajectory artefact of the 10-trajectory estimator (the exact contrast there is zero)—the sign does not invert at $n = 4$–8; the two-cell magnitude vanishes at $n = 2$ and revives at $n = 4$–8, while the three-cell contrast stays at the numerical floor.

Cells	$n = 1$	$n = 2$	$n = 4$	$n = 8$
2	$+0.0167$	-0.0001	$+0.0765$	$+0.0298$
3	-0.0021	-0.0001	≈ 0	≈ 0

size, so the single-period statement does not conceal a sustained sign flip at longer times. Second, the magnitude is not monotone in time: at two cells the contrast vanishes at $n = 2$ and revives at $n = 4$ and $n = 8$. The revival is physical, not a trajectory artefact: for the topological sector at four periods the exact Lindblad ("mesolve") integration gives a single-site negativity of 0.0754 against 0.0822 from the Monte Carlo solver on the same realization (200 trajectories), a difference of 6.8×10^{-3}. Over the full matched design the exact master-equation contrast at four periods is $+0.075\,42$ and the trajectory ladder returns $+0.0782 \pm 0.0016$, $+0.0733 \pm 0.0013$, and $+0.0751 \pm 0.0009$ at 10, 50, and 200 trajectories, while at three cells the exact contrast is zero and every ladder point is at or below 1.3×10^{-5} (Section S9 B). Monte Carlo convergence is weaker at four periods than at one: the same realization averaged over the stored 10 trajectories returns 0.0530. The revival tracks the survival of the topological-sector edge population (exact-solver edge population 0.719 at four periods versus 0.075 for the control sector, whose edge population decays toward zero under the same edge-selective rates), and it does not occur at three cells, where the contrast remains at the numerical floor from $n = 2$ to $n = 8$. The multi-period behavior therefore reinforces the finite-size, dynamical interpretation: the local contrast is a time-dependent single-period quantity whose revival, where it occurs, is tied to the engineered edge-selective profile and to coherent dynamics rather than protected ones.

I. Edge-to-bulk loss-ratio scan

The production profile sets $\gamma_{\mathrm{edge}}/\gamma_{\mathrm{bulk}} = 5 \times 10^{-4}$ ($\gamma_{\mathrm{edge}} = 10^{-4}$, $\gamma_{\mathrm{bulk}} = 0.2$). To test how strongly the positive four-mode contrast depends on this engineered sharpness, the two-cell matched design was re-run (edge-site preparation, zero disorder, 200 realizations) at boundary-to-bulk loss ratios 5×10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 5×10^{-1} and 1.0 (achieved by raising γ_{edge} at fixed $\gamma_{\mathrm{bulk}} = 0.2$):

TABLE S7. Two-cell zero-disorder edge-selective contrast versus the boundary-to-bulk loss ratio $\gamma_{\mathrm{edge}}/\gamma_{\mathrm{bulk}}$. Upper block: fixed $\gamma_{\mathrm{bulk}} = 0.2$, so the total loss budget grows with the ratio. Lower block: budget-matched scan holding $\sum_j \gamma_j = 0.4002$ (the production budget of the four-site chain), so the ratio is varied at constant total loss. The positive contrast decreases monotonically with the ratio in both scans and is lost by 5×10^{-1}.

$\gamma_{\mathrm{edge}}/\gamma_{\mathrm{bulk}}$	5×10^{-4}	10^{-3}	10^{-2}	10^{-1}	5×10^{-1}	1.0
Fixed-γ_{bulk} contrast ($d = 0$)	+0.0174	+0.0174	+0.0158	+0.0072	+0.0001	0.0000
Positive fraction	0.91	0.91	0.90	0.76	0.11	0.015
Budget-matched contrast ($d = 0$)	+0.0174	+0.0174	+0.0156	+0.0073	+0.0003	−0.0000

The positive contrast decreases monotonically with the loss ratio and is lost by 5×10^{-1}, with half of the signal already gone at 10^{-1}. Because that scan raises γ_{edge} at fixed γ_{bulk}, each ratio also changes the total loss budget. A second scan therefore holds the budget at the production value $\sum_j \gamma_j = 0.4002$ of the four-site chain—two edge sites at 10^{-4}, two bulk sites at 0.2—and distributes it over the same ratios by solving $\gamma_{\mathrm{bulk}} = 0.4002/[(n-2) + 2r]$ and $\gamma_{\mathrm{edge}} = r\,\gamma_{\mathrm{bulk}}$, on the identical disorder draws and trajectory seeds (200 realizations, $d = 0$ and 0.5). At $r = 5 \times 10^{-4}$ the two designs coincide exactly, since the production rates already satisfy the budget, and the matched scan reproduces the fixed-γ_{bulk} values at every other ratio within the paired standard errors (last row of Table S7; e.g. +0.0156 against +0.0158 at $r = 10^{-2}$ and +0.0073 against +0.0072 at $r = 10^{-1}$; at $d = 0.5$ the matched values are likewise within errors of the fixed-γ_{bulk} scan, +0.0162 at $r = 5 \times 10^{-4}$). The collapse of the signal is therefore controlled by the edge-to-bulk loss contrast rather than by the total dissipation rate: the small-system positive signal still requires the sharp edge-selective profile and should not be read as a generic feature of the dissipative chain.

J. Dimerization sweep across the gap closings

The discriminating question posed in the Introduction—whether the local contrast follows the bulk invariant, which changes at a gap closing, or the edge-mode localization, which varies continuously—is tested by scanning the intra-cell coupling along the line $J_A + J_B = 1.3$ at 19 points $J_A \in [0.1, 1.2]$, for two and three cells, with $N_{\mathrm{MC}} = 500$ realizations per point at disorder levels $d = 0$ and 0.35 pooled ($n = 1000$ matched realizations per point). All points share disorder draws and trajectory seeds, so the contrast at two neighboring points can be compared as matched pairs. The coherent invariants and quasienergy gaps reported below are evaluated on the bulk Bloch model at each point, independently of the open-system dynamics (Section S9 Q). The line contains the three expected closings: the π gap closes at $J_A = 0.15$ and $J_A = 1.15$ ($|J_A - J_B| = 1$, minimum gaps 4.4×10^{-16} and 8.9×10^{-16}) and the zero gap closes at $J_A = 0.65$ ($J_A = J_B$, minimum zero gap 2.6×10^{-16}); the bulk-point gap criterion fails at exactly these three points and passes at the other sixteen, with resolution stability at every point.

Three observations follow. First, the curve is non-monotone in the dimerization: it peaks inside the strongly dimerized topological regime ($J_A = 0.20$), is zero to within 2×10^{-5} over the window $0.6 \leq J_A \leq 0.8$ that contains the $\nu_0 = -1 \to 0$ transition and points gapped on both sides of it, and rises again in the strongly dimerized trivial regime ($J_A \approx 1.1$) before falling. Second, the endpoint differences reproduce the production sign pattern from an independent seed base: $N_{\mathrm{edge}}(0.10) - N_{\mathrm{edge}}(1.20) = +0.016\,73$ at two cells and $-0.001\,89$ at three cells, against $+0.016\,73$ and $-0.002\,10$ in the production. Third, the center-site negativity is identically zero at every point of the sweep, so the edge-minus-center contrast reduces to the edge negativity itself in this design; the column entries are therefore edge negativities rather than the matched sector contrast, which is recovered only from the endpoint difference quoted above.

To quantify whether the observable jumps at a closing, each critical point is compared with the linear interpolation of its two bracketing neighbors on the same draws. The residual is identically zero at the zero-gap closing, where the contrast itself is zero over the whole window; the largest residual anywhere is $+0.000\,86 \pm 0.000\,17$ at the low-J_A π closing (two cells), i.e. 1.95% of the local signal and below the two-cell production detectable effect $\Delta_{\mathrm{min}} = 0.001\,543$ (at $N_{\mathrm{MC}} = 500$); all other residuals are at most $0.000\,11$ in magnitude

TABLE S8. Edge-minus-center negativity contrast across the gap closings ($J_A + J_B = 1.3$, pooled over $d = 0$ and 0.35, $n = 1000$ matched realizations per point, standard errors below 7×10^{-4}; twelve of the nineteen grid points shown, full grid in the analysis report). Column entries are the edge-minus-center negativity at each dimerization point, which equals the edge negativity here because the center value vanishes identically; they are not the matched topological-minus-trivial sector contrast of main-text Table I, which is recovered only from the difference of the two endpoints. The observable is non-monotone in the dimerization, numerically zero ($\leq 2 \times 10^{-5}$) on a window that straddles the $\nu_0 = -1 \to 0$ transition, and finite again on both strongly dimerized sides.

J_A	ν_0	ν_π	Two cells	Three cells
0.10	−1	0	+0.023 50	+0.004 83
0.15	−1	−1	+0.044 33	+0.010 39
0.17	−1	−1	+0.051 19	+0.012 54
0.20	−1	−1	+0.059 28	+0.015 00
0.50	−1	−1	+0.000 03	0.000 00
0.60	−1	−1	0.000 00	0.000 00
0.65	−0.5	−0.5	0.000 00	0.000 00
0.80	0	−1	0.000 00	+0.000 02
1.00	0	−1	+0.013 36	+0.012 39
1.10	0	−1	+0.016 86	+0.016 58
1.15	0	−1	+0.012 38	+0.012 39
1.20	0	0	+0.006 78	+0.006 72

with $|z| \leq 1.32$. The contrast therefore changes continuously through all three invariant transitions. Combined with the exact-zero window on both sides of the ν_0 change, the sweep establishes that the local contrast does not track the bulk invariant in the tested model: it is a finite-localization quantity whose dependence on the dimerization is smooth and non-monotone.

That interpretation is tested directly by extracting, at each of the 19 points, the micromotion-covariant edge mode (the most edge-localized eigenvector of the one-period propagator) and measuring its localization length from the inverse participation ratio of its

site amplitudes, $\xi_{\mathrm{part}} = (\sum_j |u_j|^4)^{-1}$ in units of sites. A site-by-site exponential envelope is not identifiable on these chains: the eigenvector is a finite-size symmetric cat of the two edge profiles and changes sublattice character within the first dimer, leaving too few decay steps to fit, and the fit fails outright at and beyond the closings, where the mode is extended. The participation length is proportional to the exponential length for localized profiles, saturates at the chain length for extended ones, and is defined everywhere. As a sanity check, the deepest topological points ($J_A \leq 0.2$) are more localized than the critical window ($0.6 \leq J_A \leq 0.8$) at both sizes. Regressing the $d = 0$ contrast of the sweep against ξ_{part} over all 19 points gives Pearson $r = -0.680$ ($p = 0.0013$; Spearman $\rho = -0.583$, $p = 0.0088$; ordinary least squares $R^2 = 0.46$, slope -0.0191 ± 0.0050) at two cells, and $r = -0.312$ ($p = 0.194$, slope -0.0017) at three cells. The two-cell contrast is therefore significantly anti-correlated with the measured localization length of the edge mode; the three-cell trend has the same sign but is not resolved. The finite-localization reading is thus supported by the measured variable at the size where the contrast is significantly positive, with a directional trend at three cells.

K. Dissipation compatible with the chiral symmetry

The loss profiles used in the production do not respect the protecting symmetry: on the odd sites, $[\hat{S}, a_j] \neq 0$ with $\hat{S}$ the chiral operator. To test the symmetry-respecting row of the boundary matrix in the main text, the matched design was repeated at two and three cells ($N_{\mathrm{MC}} = 500$, $d = 0$ and 0.35, $N_{\max} = 2$, $\chi = 0.5$) with four on-site profiles. In the even-sublattice profile the loss acts only on the even (A) sublattice, so every collapse operator $L_k = \sqrt{\gamma_j} a_j$ commutes with $\hat{S}$ and the Lindbladian is strongly symmetric with respect to the chiral operator; the odd-sublattice profile damps only the odd (B) sublattice, for which $\{\hat{S}, L_k\} = 0$; the budget-matched even profile is the even profile rescaled by $n_{\mathrm{modes}}/n_{\mathrm{damped}} = 2$, so that both the total budget $\sum_j \gamma_j$ and the edge-to-bulk rate ratio match the control exactly (0.4002 at four modes and 0.8002 at six, against 0.2001 and 0.4001 for the unscaled even profile). The control reproduces the production values from an independent seed base.

Three conclusions follow. First, chiral-symmetry-compatible loss does not protect the local contrast: the strong-symmetry profile gives a small, resolved, negative contrast ($-0.000\,81 \pm$

TABLE S9. Matched local contrast under four loss profiles (mean ± standard error over $n = 500$ matched realizations per disorder level). Row labels: control is the production edge-low-loss profile; even is the even-sublattice profile with $[\hat{S}, L_k] = 0$; even (matched) is the same profile rescaled to the control budget; odd is the odd-sublattice profile. Making the dissipator commute with the chiral operator removes the positive four-mode contrast without restoring it at three cells, and matching the dissipation budget to the control does not change the outcome.

Profile	Two cells		Three cells	
	$d = 0$	$d = 0.35$	$d = 0$	$d = 0.35$
control	$+0.016\,75 \pm 0.000\,55$	$+0.016\,41 \pm 0.000\,54$	$-0.002\,14 \pm 0.000\,56$	$-0.001\,82 \pm 0.000\,53$
even	$-0.000\,81 \pm 0.000\,05$	$-0.000\,78 \pm 0.000\,04$	$-0.000\,80 \pm 0.000\,05$	$-0.000\,77 \pm 0.000\,04$
even (matched)	$-0.000\,79 \pm 0.000\,09$	$-0.000\,85 \pm 0.000\,10$	$-0.000\,78 \pm 0.000\,09$	$-0.000\,84 \pm 0.000\,10$
odd	$-0.001\,80 \pm 0.000\,52$	$-0.002\,32 \pm 0.000\,57$	$-0.001\,80 \pm 0.000\,52$	$-0.002\,32 \pm 0.000\,57$

0.000 05 at two cells, $d = 0$), a twenty-fold reduction in magnitude relative to the control and a sign opposite to the four-mode production sign. Second, this is not a loss-budget artefact: the budget-matched profile ($-0.000\,79 \pm 0.000\,09$) is statistically indistinguishable from the half-budget one, so the result holds at fixed total dissipation and fixed edge-to-bulk ratio. Third, the mechanism differs by profile: under the strong-symmetry profile both sectors retain nearly the same edge negativity (topological 0.022 85, trivial 0.023 66 at two cells, $d = 0$), so the profile suppresses the sector distinction rather than the resource, whereas the odd-sublattice profile suppresses the resource in both sectors (0.004 49 and 0.006 29). The sublattice-resolved profiles are also nearly size-independent (mean paired four-versus-six-mode difference 3.9×10^{-6} for the even profile, against 9.5×10^{-3} for the control), because the strongly damped sites decouple the edge site from the far end of the chain. This test implements on-site sublattice-localized loss, i.e. $[\hat{S}, L_k] = 0$; other strong-symmetry classes (pair loss, $U(1)$ or parity symmetry) are outside its scope.

L. Sub-period dependence of the contrast

The production observable is evaluated after one full drive period. To test how strongly the value depends on the measurement phase within the drive period, the same matched

protocol was evaluated at the nine equispaced times $t/T \in \{0, 0.125, \ldots, 1\}$ for two and three cells, $N_{\mathrm{MC}} = 500$, $d = 0$ and 0.35, under edge-selective and uniform loss ($N_{\max} = 2$, $\chi = 0.5$, 10 trajectories). The stored series reproduces the terminal observable exactly, so the $t = T$ column follows the production convention on an independent seed base.

TABLE S10. Matched contrast (topological minus trivial edge negativity) as a function of the phase of the drive, at zero disorder. The contrast changes sign within the period under every tested profile and size; the uniform-loss null of the production is a statement about $t = T$ only.

t/T	2 cells, edge-selective	3 cells, edge-selective	2 cells, uniform
0	0.0000	0.0000	0.0000
0.125	+0.1922	+0.1922	+0.1090
0.25	+0.1854	+0.1854	+0.0499
0.375	+0.1339	+0.1339	+0.0094
0.5	+0.1667	+0.1667	+0.0083
0.625	−0.0070	−0.0070	−0.0001
0.75	−0.0070	−0.0072	−0.0001
0.875	+0.1021	+0.0269	+0.0003
1	+0.0167	−0.0021	-8.1×10^{-7}

Four observations follow. The contrast is identically zero at $t = 0$, where both sectors share the same prepared state, and the $t = T$ column reproduces the production values (+0.0167 and −0.0021). The intra-period swing is an order of magnitude larger than the stroboscopic contrast and the sign changes inside every one of the eight tested conditions (two sizes, two profiles, two disorder levels); for two cells, edge-selective loss, $d = 0$, the matched difference at $5T/8$ is $-0.007\,02 \pm 0.000\,41$. The sector-resolved curves show why: the trivial sector's local negativity collapses to ≈0.007 within $T/8$ and stays there, while the topological sector holds ≈0.17 until $T/2$, falls to an exact grid zero at $5T/8$ and $3T/4$, and partially revives by $7T/8$; the two sizes are indistinguishable ($< 10^{-7}$) up to $T/2$ and separate by 0.075 at $7T/8$, consistent with the arrival of the chain-boundary difference at the edge site. Finally, under uniform loss the stroboscopic null is not an intra-period statement: the contrast reaches +0.1090 at $T/8$ before returning to ≈0 at T. The reported contrast is therefore a stroboscopic, phase-of-the-drive conditional quantity, which is why the claim boundary

of the main text fixes the measurement at $t = T$ consistent with micromotion-covariant Floquet accounting [7].

M. Out-of-sample five-cell size

The size dependence of the main text rests on two, three and four unit cells. A five-cell extension (ten modes, Hilbert space $3^{10} = 59\,049$) was run under the matched production design with $N_{\mathrm{MC}} = 300$, $d = 0$ and 0.5, both profiles, both sectors and the production seed base (2400 records). Before the run was submitted, the prediction for this size was frozen: an exponential-saturation model and an inverse-size model were fitted on the two-to-four-cell production means, combined into an ensemble envelope of $\pm 1.96\,\mathrm{SE}(300)$, and recorded as the frozen prediction before the run. The measured contrast is scored against that frozen prediction.

TABLE S11. Measured five-cell contrast versus the frozen out-of-sample prediction (edge-minus-center negativity contrast, topological minus trivial, paired draws; SE is the standard error of the paired mean, CI the normal-theory 95 % interval, and the frozen CI the prediction interval written before the run). Every measured mean falls inside the frozen interval (4 of 4 conditions); the two sign-resolved edge-selective conditions match the frozen sign prediction and the two uniform-loss conditions are sign-unresolved, as predicted.

Statistic	$d = 0$, edge	$d = 0$, uniform	$d = 0.5$, edge	$d = 0.5$, uniform
Measured mean	−0.002 176	−0.000 001	−0.002 606	+0.000 029
SE	0.000 737	0.000 005	0.000 720	0.000 027
CI lower	−0.003 621	−0.000 012	−0.004 017	−0.000 024
CI upper	−0.000 732	0.000 010	−0.001 194	0.000 082
Frozen lower	−0.010 240	−0.000 011	−0.010 199	−0.000 071
Frozen upper	−0.000 712	0.000 009	−0.000 893	0.000 068
Sign	negative	unresolved	negative	unresolved

All four conditions lie *inside* the frozen intervals; the two edge-selective conditions match the frozen sign and the two uniform-loss conditions are sign-unresolved, exactly as frozen. The exponential-saturation point prediction is nearly exact (−0.002 116 predicted versus

$-0.002\,176$ measured at $d = 0$, a gap of 0.08 standard errors; $-0.002\,324$ versus $-0.002\,606$ at $d = 0.5$), while the inverse-size model overestimates the magnitude by roughly a factor of four (-0.0088), lying outside the measured 95 % interval in both edge-selective conditions. The frozen ensemble band includes the inverse-size model by construction (it is an envelope over both registered models), so the exclusion of the $1/N$ drift comes from the measured intervals and not from the frozen prediction itself; the frozen ensemble central value $-0.005\,476$ misses the measured $d = 0$ mean by 4.5 standard errors. The four-size trend at $d = 0$, edge-selective loss is $+0.016\,731 \pm 0.000\,550$ (two cells), $-0.002\,104 \pm 0.000\,557$ (three), $-0.002\,009 \pm 0.000\,273$ (four, $N_{\mathrm{MC}} = 2000$; the frozen fits used the then-current $N_{\mathrm{MC}} = 500$ value $-0.002\,110 \pm 0.000\,558$), $-0.002\,176 \pm 0.000\,737$ (five): the sign change between two and three cells persists at five cells and the negative branch saturates at magnitude ≈ 0.0021 rather than drifting as $1/N$ (the measured intervals exclude the inverse-size point predictions).

N. Grid-free witness of Wigner negativity

The production negativity is a discretised integral on a 31×31 Wigner grid. To certify that the two-cell signal does not rest on that discretisation, the complete production design was rerun with terminal 3×3 reduced edge and center blocks recorded (8000 records per size). The rerun is gated on exact equality: every record reproduces the archived production edge and center negativities, rates, and draw and trajectory seeds (24 000/24 000 pass), and every block must be Hermitian, unit-trace and positive semidefinite. On that identical trajectory ensemble, the analysis evaluates the Fock-fidelity witnesses of Wigner negativity introduced by Chabaud *et al.* [4], $\hat{\Omega}_{n,\alpha} = D(\alpha)|n\rangle\langle n|D^{\dagger}(\alpha)$: if $\max_{\alpha}\langle n|D^{\dagger}\rho D|n\rangle > \omega_n$ then the state has a negative Wigner function. The threshold does not depend on α because Wigner negativity is displacement invariant, so the maximisation over α (coarse 81×81 grid plus a 0.0025 refinement) is part of the witness rather than a tuning choice, so the reported certification rates are conservative lower bounds of the witness functional; $\omega_1 = \omega_2 = 0.5$ are the analytical values and $n = 3, \ldots, 6$ use the published upper bounds 0.427/0.441/0.385/0.378 conservatively. The α search itself is converged: re-maximising the certified $n = 1$ edge witness on every one of the 24 000 stored blocks with a dense 161×161 grid and a 0.0005 refinement around the three largest coarse maxima moves any reported F_1 by at most 2.4×10^{-6}

(stratum-level mean shifts and paired ΔF_1 shifts below 10^{-6}) and flips no certification in any of the 24 strata; across all six witness indices and both sites the dense search changes any fidelity by at most 4.3×10^{-4}, and the centre-site control crosses no threshold at any n, so the reported rates are not search-resolution limited. As a basis-independent companion the relative-entropy non-Gaussianity [8] $\Delta = S(\rho_{\mathrm{G}}) - S(\rho)$ is computed with moments taken in an embedded $d + 4$ space so that truncated-operator boundary terms do not corrupt the covariance matrix (validation: the Fock state $|2\rangle$ reproduces the analytic value $\Delta = 3\ln 3 - 2\ln 2 = 1.910$; coherent states are Gaussian and give $\Delta = 0$ exactly).

TABLE S12. Edge-site witness statistics at $d = 0$, edge-selective loss, from the production rerun gated on exact equality. Certified: fraction of disorder draws with $\max_\alpha F_1 > \omega_1 = 0.5$ (Wilson intervals [9] for the headline two-cell rates: topological [0.788, 0.855], trivial [0.084, 0.138], disjoint). Under uniform loss no draw certifies at any size or sector, and the center-site control equals the vacuum value $\max_\alpha F_1 = e^{-1} = 0.367\,879$ at two and four cells (all 500 draws) and never exceeds 0.416 at three cells.

	Topological			Trivial		
Size	F_1^*	certified	Δ	F_1^*	certified	Δ
2 cells	0.4950	412/500	0.3355	0.4529	54/500	0.2039
3 cells	0.4481	57/500	0.2045	0.4525	50/500	0.2053
4 cells	0.4481	57/500	0.2045	0.4525	50/500	0.2054

At two cells the topological sector certifies in 82.4 percent of disorder draws against 10.8 percent for the trivial sector (paired F_1 difference $+0.042\,15 \pm 0.002\,00$, 21 standard errors), and the paired non-Gaussianity difference is $+0.1316 \pm 0.0025$. Under edge-selective loss the certification rate degrades with disorder (412, 291, 242, 226 of 500 for $d = 0$, 0.5, 1, 1.5) while the paired difference stays resolved at ≥ 17 standard errors. At three and four cells neither sector certifies preferentially (11.4 versus 10.0 percent; paired F_1 difference $-0.004\,46 \pm 0.002\,41$), the witness equivalent of the small negative grid contrast, and the aggregate values at the two sizes agree to $\sim 2 \times 10^{-6}$ because the edge-triad state at $t = T$ is insensitive to chain length beyond three cells on the shared disorder draws. The witness therefore certifies the two-cell Wigner negativity grid-free on the identical production trajectories while returning null results for the uniform-loss control, the center site, and the three- and

four-cell states.

O. Long-time evolution to sixty-four periods

The production observable is a single-period snapshot. To test whether the sign pattern survives the approach to the stationary state, the matched protocol was propagated on the stroboscopic grid $t_k = kT$ for $k = 0, \ldots, 64$ at two and three cells, $N_{\mathrm{MC}} = 200$, all four disorder levels, both profiles and sectors, with the production propagator and seed base (3200 records per size). All sixteen strata reproduce the archived production contrast at $t = T$ within three combined standard errors (largest deviation 6.3×10^{-4}), so the curves start from the production snapshot.

TABLE S13. Paired edge-negativity contrast at selected stroboscopic times for $d = 0$, edge-selective loss (mean ± standard error over matched draws; zero denotes the numerical grid floor). The two-cell branch vanishes at $2T$, reaches its largest value at $4T$, and decays to the floor by $\approx 37T$; the three-cell branch decays monotonically to the floor by $6T$ and never becomes significantly positive at any of the 64 periods.

$k = t_k/T$	2 cells	3 cells
1	$+0.017\,140 \pm 0.000\,873$	$-0.001\,887 \pm 0.000\,845$
2	$-0.000\,099 \pm 0.000\,035$	$-0.000\,110 \pm 0.000\,039$
3	$+0.019\,534 \pm 0.000\,922$	$-0.000\,030 \pm 0.000\,012$
4	$+0.077\,204 \pm 0.002\,603$	$-0.000\,009 \pm 0.000\,003$
6	0	0
8	$+0.030\,278 \pm 0.001\,753$	0
12	$+0.015\,239 \pm 0.001\,252$	0
16	$+0.006\,744 \pm 0.000\,793$	0
24	$+0.000\,566 \pm 0.000\,139$	0
32	$+0.000\,062 \pm 0.000\,026$	0
64	0	0

At two cells the sign at T persists at every remaining sampled period after the small estimator-level reversal at $2T$, which a trajectory ladder against the exact master equation

resolves as a finite-trajectory artefact (the $2T$ exact contrast is zero; Section S9 B), which is the multi-period vanishing already documented in Section S9 H; the post-revival decay is monotone from $8T$ to the floor, reached at $k \approx 37$ (the $k = 6$ zero is a below-floor censored entry). At $d = 1$ and 1.5 the plateau is stronger (median over $4T$–$16T$: $+0.012\,90 \pm 0.001\,05$ and $+0.010\,71 \pm 0.001\,36$) and no stationary state is reached within the 64-period horizon. At three cells the contrast decays monotonically to the floor by $6T$ ($d = 0$) or $5T$ ($d = 0.5$) and by $27T$ at $d = 1$, where an envelope fit gives an effective decay time $\tau \approx 7.5$ periods; for $d \leq 1$ the three-cell contrast never becomes significantly positive at any of the 64 periods, so the production sign is preserved throughout the decay. The uniform-loss null is stationary from $k = 1$ at $d \leq 0.5$ ($|\Delta| \leq 3 \times 10^{-5}$). One caveat appears at the strongest disorder: at $d = 1.5$, three cells, a negative contrast grows at long times in *both* profiles (at $64T$: $-0.001\,588 \pm 0.000\,262$ edge-selective, $-0.001\,195 \pm 0.000\,233$ uniform), so at $d = 1.5$ the uniform-loss null is a statement about $t = T$ only. Both size branches are therefore consistent with decay-to-steady-state dynamics in which the sign survives the decay but no persistent protected resource remains.

P. Collective-mode Wigner observable

The local observable of the main text is the single-site Wigner negativity Eq. (S7). A collective-mode statement requires a bosonic mode transform: for each sector we take the normalized single-particle edge vector $c = (c_0, \ldots, c_{N-1})$ (the eigenvector of the sector Floquet operator with maximal edge weight, $\sum_j |c_j|^2 = 1$) and define the collective edge mode $b_0 = \sum_j c_j a_j$. A unitary $N \times N$ mode rotation U with first row c is constructed deterministically [10], and the exact many-body rotation $\mathcal{U}$ on the truncated Fock space is built by acting with the rotated creation operators on the vacuum,

$$\mathcal{U}|n_1 \ldots n_N\rangle = \prod_j \frac{1}{\sqrt{n_j!}} \left(\sum_i u_{ji}\, a_i^\dagger \right)^{n_j} |0\rangle, \tag{S15}$$

after which the collective-mode marginal is $\rho_{\mathrm{edge}} = \mathrm{Tr}_{j \neq 0}\, \mathcal{U} \rho \mathcal{U}^\dagger$ and its Wigner negativity is evaluated by Eq. (S7).

We verified the mode transform before interpreting the collective observable. (i) Single-particle unitarity of U holds to 3×10^{-15}, and the many-body rotation Eq. (S15) is unitary to machine precision on the total-excitation sector $\leq N_{\max}$ (Frobenius defect 1.0×10^{-14}–

4.4×10^{-14} for $N_{\max} = 2, 3, 4$); the particle-number conserving Hamiltonian and loss leave the dynamics inside that sector. (ii) A coherent excitation placed on the collective edge mode yields collective negativities 0.0973, 0.0546, 0.0234 at $N_{\max} = 2, 3, 4$, reproducing the single-mode truncation-artifact values exactly—the rotated-frame Gaussian control behaves identically to the local one. (iii) A cat excitation on the edge mode reproduces the single-mode cat negative volume to a relative deviation of 3×10^{-15} at all tested truncations, and the identity rotation reproduces the local observable exactly (maximum absolute difference 0.0). Using the unitary transform, the same matched conditions analyzed for the local observable were re-analyzed for the collective edge-mode marginal at both production-relevant Fock truncations $N_{\max} = 2$ and $N_{\max} = 3$ (two cells, four modes, both sectors, edge-selective and uniform loss, disorder $d = 0$ and 0.35, 10 matched disorder realizations, 10 trajectories each, 80 records per truncation, 160 records in total). The collective edge-mode Wigner negativity is not resolved above the numerical floor in any of the 160 records (maximum value 0.0) at both truncations, in both loss profiles and both sectors, whereas the local site-0 negativity is positive under edge-selective loss at both truncations (paired chiral-minus-trivial contrast 0.0175 ± 0.0040 at $N_{\max} = 2$, 90 % of realizations positive at $d = 0$; 0.0105 ± 0.0033 at $N_{\max} = 3$, 70 % positive at $d = 0$) and vanishes under uniform loss—the same result. The null result also holds at larger system size: an independent 80-record analysis at three cells / six modes / $N_{\max} = 2$ (same design, 10 matched realizations) returns no resolved collective edge-mode negativity in any of the 80 records, with the local contrast consistent with the size-sensitivity of the main text (at three cells the paired chiral-minus-trivial local contrast is 0.0003 ± 0.0041 at $d = 0$, i.e. the edge-selective local advantage of the two-cell chain does not persist at three cells, matching the size-sensitivity reported in Figs. 2 and 4). The truncation-stability extension at three cells / six modes / $N_{\max} = 3$ (Fock-space dimension 4096) confirms the same verdict: an independent 80-record analysis with the identical matched design returns no resolved collective edge-mode negativity above the tested numerical floor in any of the 80 records, with the identity-rotation consistency check exact, and a paired local contrast of -0.0017 ± 0.0033 at $d = 0$ (statistically null, matching the three-cell size-sensitivity). The absence of resolved collective negativity above the tested numerical floor is therefore stable across both truncations and both system sizes (320 records in total); this is a finite-grid, finite-cutoff numerical null rather than an exact-zero theorem. The local-contrast magnitude is, as noted above, not cutoff-converged (it varies by roughly

a factor of two between the two truncations at two cells), but its sign and positivity under edge-selective loss, and the absence of resolved collective negativity, are truncation-stable. The absence of resolved negativity is not explained by the tested grid or truncation: for the representative topological state the edge-mode Wigner function is positive everywhere, with minimum 2.3×10^{-13} at $N_{\max} = 2$ and 1.5×10^{-13} at $N_{\max} = 3$, unchanged across 31×31, 61×61 and 91×91 grids, while the local marginal retains a genuine negative region (Wigner minimum -0.039 at $N_{\max} = 2$ and -0.026 at $N_{\max} = 3$); both marginals are shown in Fig. 3. The same collective-mode test was then applied to the *mode-resolved* preparation (the topological edge-mode cat, "topological zero mode"), which had not previously been submitted to the collective Wigner analysis. With that preparation the local site-0 marginal is itself Gaussian from the start (the edge mode has boundary weight $|u_{e,0}|^2 = 0.013$ and is 0.99 delocalized), and the 240 additional records (two cells at $N_{\max} = 2$ and 3, three cells at $N_{\max} = 2$; both sectors, both loss profiles, $d = 0$ and 0.35) return no resolved local site-0 negativity, and collective edge-mode negativity at or below the truncation floor (maximum collective value 1.2×10^{-5} at two cells / $N_{\max} = 3$; no resolved value elsewhere), with the identity-rotation consistency check exact. Combined with the common-site results above, the collective Wigner null now covers both preparations, both loss profiles, two system sizes and two Fock truncations (560 records in total): no collective edge-mode Wigner negativity is resolved above the tested numerical floor under either preparation.

The collective-mode analysis gives a numerical null result: the unitary transform resolves no collective edge-mode negativity above the reported numerical floor in any matched condition, consistent with and reinforcing the size- and profile-sensitivity findings of the main text. Local negativity must not be described as collective-mode negativity—the two observables differ, and only the local one is nonzero here.

Q. Spectral-localizer analysis of the coherent sectors

As an independent calculation of the coherent-model sectors, we implement the spectral localizer for the two symmetric-time-frame unitaries of the chiral chain ($U_A = bab$, $U_B = aba$; the frames whose off-diagonal phase windings define w_A, w_B in the main text). For each frame the effective Hamiltonian $H_{\text{eff}}^{A,B} = -i \ln U_{A,B}$ satisfies the chiral relation $S H_{\text{eff}}^{A,B} S = -H_{\text{eff}}^{A,B}$ with the chiral operator $S = \mathrm{diag}(1, -1, 1, -1, \dots)$, and the local winding marker is

the half-signature of the symmetry-reduced spectral localizer [11, 12]

$$\nu_{A,B}(x) = \tfrac{1}{2}\,\mathrm{sig}\big[\big(\kappa(X-x) - iH_{\mathrm{eff}}^{A,B}\big)S\big], \qquad W_0 = \tfrac{1}{2}(\nu_A+\nu_B), \quad W_\pi = \tfrac{1}{2}(\nu_A-\nu_B), \tag{S16}$$

with X the per-cell position operator and κ the localizer scaling constant. The marker is constant through the bulk and the localizer gap $\sigma_L(x,0) = \min|\sigma[\tilde{L}_{(x,0)}]|$ collapses at the boundaries precisely where boundary modes sit, the "local topological bandgap" criterion.

The spectral-localizer calculation gives the following results:

- *Topological point* (J_A/J_B of the main-text nontrivial sector): the localizer markers are $W_0 = 1$, $W_\pi = 0$ for $N = 4, 6, 8$ cells, reproducing the classification of the frame windings $(\nu_0, \nu_\pi) = (-1, 0)$ up to the k-orientation sign convention. The localizer gap profile at $N = 8$ is $(4\times10^{-4}, 0.148, 0.295, 0.316, 0.316, 0.295, 0.148, 4\times10^{-4})$, i.e., it closes at the two boundaries and stays finite in the bulk (Fig. S3(a)–(b)).

- *Trivial point*: $W_0 = W_\pi = 0$ and a flat gap (≈ 0.35 across all cells), i.e., no boundary modes.

- *Quantitative protection bound (Lemma).* For the two-step drive with on-site disorder of amplitude d added to both step Hamiltonians (entries i.i.d. uniform $[-1,1]$), the accumulated instantaneous-Hamiltonian disorder is

$$D_{\mathrm{acc}} \equiv \frac{1}{T}\int_0^T \|\delta H(t)\|\,\mathrm{d}t = d\,\frac{\|D_a\| + \|D_b\|}{2} \le d, \tag{S17}$$

  and the index-relevant bulk localizer gap obeys the linear floor

$$\sigma_L^{\mathrm{bulk}}(d) \;\ge\; \sigma_L^{\mathrm{bulk}}(0) - d + O(d^2), \qquad \sigma_L^{\mathrm{bulk}}(d) \equiv \min_{\substack{x\in\mathrm{bulk}\\ E\in\{0,\pi/T\}}} \sigma_L(x, E; H_{\mathrm{eff}}(d)), \tag{S18}$$

  where the boundary cells are excluded from the minimum: topological edge modes legitimately close the localizer gap there (the "local topological bandgap" criterion), and the marker is evaluated in the bulk. The linear floor is the first-order Wong–Cerjan–Cole bound on the perturbation of H_{eff} [13]; the $O(d^2)$ remainder arises from the nested commutators of the Magnus expansion. While the floor stays positive the sector is pinned by homotopy: a sector transition requires the bulk localizer gap to close. We emphasize that this on-site potential perturbation of the *coherent* model is a different noise model from the multiplicative rate noise used in the open-system

production (Section S3); the bound above therefore certifies the coherent sector under its own perturbation model and is not a disorder-response statement for the Lindblad dynamics, even though both are denoted by d.

- *Numerical verification of the bound* (20 realizations per point and size, $d \in \{0, 0.05, 0.1, 0.2, 0.35, 0.5\}$; Fig. S2): the bound Eq. (S18) holds for every realization at every level for both points at all three bulk sizes ($N = 6, 8, 10$ cells), with minimum margins to the linear floor 0.035 (topological, $N = 6$, $d = 0.05$) and 0.020 (trivial, $N = 6$, $d = 0.05$); the margins grow with d because the measured degradation is sub-linear (mean-bulk-gap slopes -0.083 and -0.086 at $N = 6$, -0.084 and -0.071 at $N = 8$, -0.095 and -0.108 at $N = 10$, versus the unit-slope floor; Fig. S3(c) shows the $N = 6$ profiles). The bound certifies sector stability for all $d < d^* \equiv \sigma_L^{\mathrm{bulk}}(0)$; d^* is size-stable, 0.138/0.108/0.096 (topological, $N = 6/8/10$) and 0.102/0.096/0.090 (trivial, $N = 6/8/10$), converging toward the thermodynamic bulk gap. The measured bulk gap stays open at all tested levels at all three sizes (worst case $\min \sigma_L^{\mathrm{bulk}} \geq 0.029$ topological at $N = 8$, ≥ 0.017 at $N = 10$; ≥ 0.005 trivial at $N = 6$, ≥ 0.004 at $N = 10$, at $d = 0.5$), so no sector transition occurs in the tested range—the trivial point's gap approaching closure at $d = 0.5$ is consistent with its smaller d^* and is likewise size-stable.

- *Flip-detection caveat.* On-site diagonal disorder commutes with the chiral operator ($SDS = D$), so $S(H + D)S = -H + D \neq -(H + D)$: chiral symmetry is broken at $d > 0$, and the reduced half-signature markers Eq. (S16) (which require exact chiral symmetry) lose their pinning—their fluctuation under disorder (up to 0.35 of realizations for the topological point at $d \geq 0.35$) is a symmetry-breaking artifact of the marker formula, not a sector transition. The valid sector statement is therefore gap-based: the sector is pinned while the bulk localizer gap is open, which is exactly the content of the bound Eq. (S18).

This provides the required independent recomputation: the sector classification is reproduced by a second, structurally different implementation (half-signature of the reduced localizer vs. off-diagonal phase winding), with disorder and finite-size controls. The bound Eq. (S18) is a quantitative coherent-model stability statement, not a protection statement for the open-system Wigner observable: the localizer gap degrades at most linearly in the

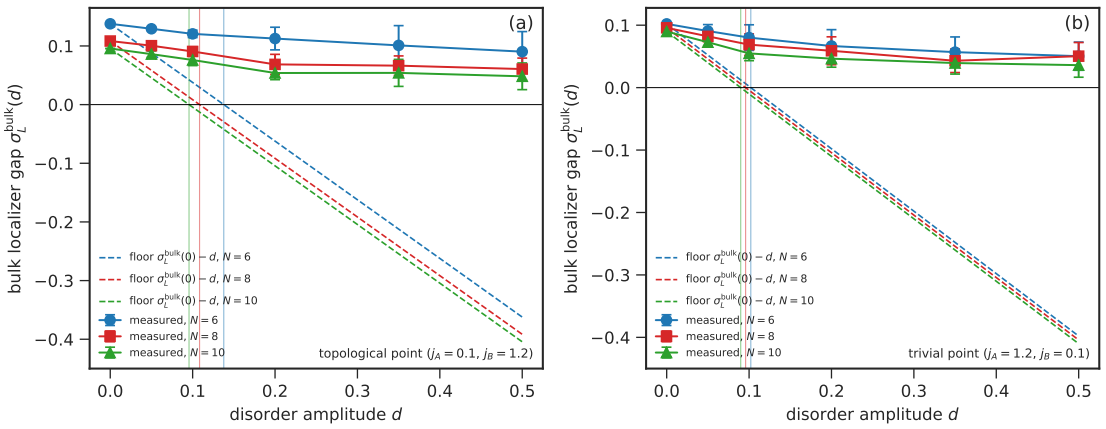


FIG. S2. Disorder resilience of the protection bound Eq. (S18) at three bulk sizes ($N = 6, 8, 10$ cells), for the (a) topological and (b) trivial points. Points: measured index-relevant bulk localizer gap $\sigma_L^{\mathrm{bulk}}(d) = \min_{x \in \mathrm{bulk}, E \in \{0, \pi/T\}} \sigma_L(x, E)$ (mean over 20 disorder realizations, error bars one standard deviation). Dashed lines: the linear floor $\sigma_L^{\mathrm{bulk}}(0) - d$ of the lemma, whose zero crossing marks the certified stability limit d^*. The measured gap stays strictly above its floor at every level for both points and all three sizes (sub-linear degradation), and never closes up to $d = 0.5$: the sector is pinned over the entire tested range, and the floor itself is size-stable (d^*: 0.138/0.108/0.096 topological, 0.102/0.096/0.090 trivial for $N = 6/8/10$). Vertical ticks mark d^* at each size.

accumulated instantaneous-Hamiltonian disorder, and the sector is pinned while the gap is open—verified numerically above. It is a protection statement for the *coherent* chiral chain only and does not extend to the open-system resource (Wigner negativity), whose protection is assessed separately in the main text.

R. Transfer analysis: when does coherent protection reach the open-system observable?

The protection bound Eq. (S18) constrains the single-particle spectrum of the coherent chain. The open-system observable of the main text, the local Wigner negativity $\delta_-(t)$ of the edge site, is a many-body functional of the full dissipative trajectory and is generated at the edge site by the Kerr nonlinearity $H_{\mathrm{K}} = \chi\, a_1^{\dagger 2} a_1^2$. The transfer of coherent protection to this functional requires conditions beyond what the localizer gap controls.

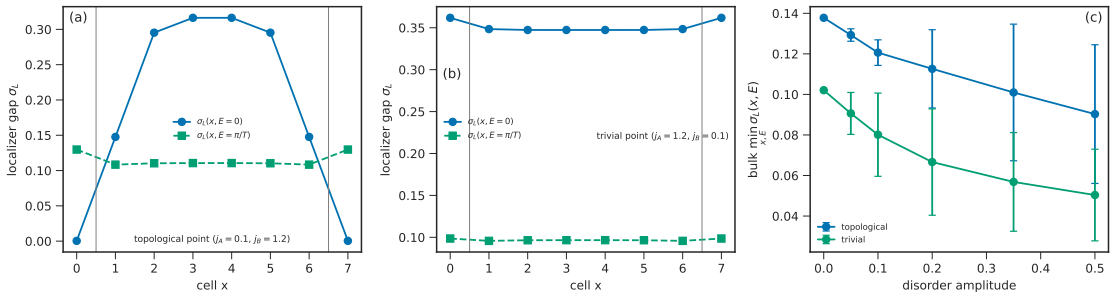


FIG. S3. Spectral-localizer certification profiles. (a) Localizer gap $\sigma_L(x, E)$ versus cell x at $N = 8$ cells for the topological point, and (b) the same profile for the trivial point, at quasienergies $E = 0$ and $E = \pi/T$. For the topological point the gap collapses at the two boundary cells (the "local topological bandgap" criterion, Eq. (S16)), whereas the trivial point shows a flat, gapped profile. (c) Index-relevant bulk gap $\min_{x,E} \sigma_L(x, E)$ versus disorder amplitude ($N = 6$ cells, 20 realizations): the bulk gap degrades sub-linearly and stays open up to $d = 0.5$, consistent with the protection bound Eq. (S18).

a. Lemma (bulk-participation decay bound). Let $|e\rangle = \sum_j u_{e,j} |j\rangle$ be the single-particle edge mode and let the edge site be $j = 0$. The local negativity $\delta_-(t)$ satisfies the short-time bound

$$\delta_-(t) \;\le\; \delta_-(0)\, e^{-\Gamma_{\text{eff}} t}, \qquad \Gamma_{\text{eff}} \;=\; \gamma_0 |u_{e,0}|^2 + \sum_{j \neq 0} \gamma_j |u_{e,j}|^2 + \gamma_{\text{eb}}, \tag{S19}$$

where γ_j are the per-site loss rates and γ_{eb} is the edge–bulk dephasing rate generated by the mode delocalization. Two consequences follow. First, edge-selective loss ($\gamma_0 \to 0$) removes only the first term; the residual decay $\sum_{j \neq 0} \gamma_j |u_{e,j}|^2 + \gamma_{\text{eb}}$ survives and grows with the bulk participation of the edge mode. Second, as the chain grows the boundary weight $|u_{e,0}|^2$ decreases and the local reduced state is averaged over an increasing number of weakly occupied sites, pushing $W(q, p)$ toward a Gaussian form—a mechanism consistent with the observed sign change of the chiral-minus-trivial contrast between two, three and four cells (main text, Fig. 2); the sign change itself is measured, not derived from this lemma.

We verified Eq. (S19) directly against the chiral production data (matched-mean final negativity; $\delta_-(0) = 0.1962$ from the identical cat-state edge-site initial condition on the same 31×31 Wigner grid). For the edge-low-loss profile at zero disorder, the measured decay rate $\Gamma_{\text{meas}} = -\ln[\delta_-(T)/\delta_-(0)]/T$ exceeds the mode-weighted rate $\Gamma_{\text{eff}} = 0.195$ at every size — 0.337 (2 cells), 0.591 (3 cells), 0.591 (4 cells) — so the bound holds with

TABLE S14. Verification of the bulk-participation decay bound (S19) against the chiral production data (edge-low-loss profile, zero disorder): the measured decay rate always exceeds the mode-weighted rate, with margins 1.7–3.0×, and the residual grows with size. Conditions and definitions are given in the text.

Cells	$\lvert u_{e,0}\rvert^2$	Γ_{eff}	$\delta_-(T)$	$\delta_-(0)$	Γ_{meas}	$\Gamma_{\mathrm{meas}}/\Gamma_{\mathrm{eff}}$
2 ($N_{\mathrm{MC}} = 500$)	0.013 43	0.1946	0.023 69	0.196 23	0.3365	1.73
3 ($N_{\mathrm{MC}} = 500$)	0.012 21	0.1951	0.004 78	0.196 23	0.5913	3.03
4 ($N_{\mathrm{MC}} = 500$)	0.012 20	0.1951	0.004 78	0.196 23	0.5913	3.03

margins 1.7–3.0×. The residual $\Gamma_{\mathrm{meas}} - \Gamma_{\mathrm{eff}}$ (the effective γ_{eb}) grows from 0.14 to 0.40 as the chain grows, quantitatively consistent with the delocalization mechanism: the bulk participation of the edge mode, weighted by the bulk loss rates, dominates the early decay of the local negativity. The full verification record is compiled in Table S14.

Extending the check to *every* realization of both loss profiles and all four disorder levels (3 sizes × 2 profiles × 4 levels = 24 conditions, 12 000 records, using each record's own disorder-drawn rates), the mean-level ratio $\Gamma_{\mathrm{meas}}/\Gamma_{\mathrm{eff}}$ is between 1.4 and 8.4 in all 24 conditions, and per-realization compliance is 100 % in 20/24 conditions (uniform loss: 100 % everywhere). The only sub-100 % cases are edge-low-loss at disorder $d = 1.0$–1.5 (compliance 77.4 %–99.4 %), where the ±100 %–150 % rate perturbation drives some realizations' bulk rates far above the mean (Γ_{eff} reaches 0.39 at $d = 1.0$ and 0.49 at $d = 1.5$, compared with ≈ 0.31 and ≈ 0.31–0.45 for the failing realizations, respectively); there the fixed-mode assumption of the bound — clean-model mode weights with given rates — is stressed by the disorder itself, a documented regime boundary rather than a violation of the mean-level statement. At the largest amplitude the same clamping also zeroes *every* rate in 14 %–16 % of the draws; those realizations evolve without dissipation, the bound then reduces to $\delta_-(T) \le \delta_-(0)$, and they are tested against—and satisfy—that inequality rather than being excluded from the compliance count. The per-condition compliance record is compiled in Table S15.

b. Closed form of the edge–bulk rate. The residual $\gamma_{\mathrm{eb}} = \Gamma_{\mathrm{meas}} - \Gamma_{\mathrm{eff}}$ can be computed in closed form from the coherent model, without fitting. The local negativity is a two-photon (cat) coherence of the edge site, so its loss exposure is set not by the single-particle

TABLE S15. Extended verification of the bound (S19) per realization over the full disorder grid: 3 sizes × 2 profiles × 4 levels = 24 conditions (12 000 records), each record's own disorder-drawn rates. Mean-level ratio $\Gamma_{\text{meas}}/\Gamma_{\text{eff}}$ (valid-only condition-level ratio: Γ_{meas} evaluated at the mean final negativity over the realizations with $\delta_-(T) > 0$, divided by their mean Γ_{eff}) and per-realization compliance $\Gamma_{\text{meas}} \geq \Gamma_{\text{eff}}$. The only sub-100 % conditions are edge-low-loss at $d = 1.0$–1.5.

Profile	d	$\Gamma_{\text{meas}}/\Gamma_{\text{eff}}$ (range)	Compliance	# conditions
edge-low-loss	0.0	1.73–2.71	100 %	3
edge-low-loss	0.5	1.73–2.80	100 %	3
edge-low-loss	1.0	1.72–3.04	86.4 % (2c); 100 % (3c, 4c)	3
edge-low-loss	1.5	1.38–2.71	77.4 % (2c); 99.4 % (3c); 99.4 % (4c)	3
uniform	0.0	5.26	100 %	3
uniform	0.5	5.43	100 %	3
uniform	1.0	8.41	100 %	3
uniform	1.5	6.84	100 %	3

population but by the *rate-weighted dwell time of the two-photon sector*:

$$\gamma_{\text{eb}} \;=\; \kappa\,\Lambda_2^{\text{coh}}, \qquad \Lambda_2^{\text{coh}} \;=\; \frac{1}{T}\int_0^T dt \sum_{\substack{i\le j\\ (i,j)\neq(0,0)}} (\gamma_i+\gamma_j)\,\big|\langle i,j\,|\,U_2(t)\,|0,0\rangle\big|^2, \tag{S20}$$

where $U_2(t)$ is the two-particle Floquet evolution of the coherent two-step chain and $|0,0\rangle$ the two-photon state at the edge site. The prefactor $\kappa = \gamma_{\text{eb}}^{\text{meas}}/\Lambda_2^{\text{coh}}$ is fixed per size by the measured residual, not fitted (Table S16); the closed-form decoherence constant of the bare truncated cat is κ_{single}, derived below. Photon loss on a bare site is a Gaussian channel; its Wigner function evolves by the exact damped-oscillator solution

$$W_t(\boldsymbol{\beta}) \;=\; \frac{2}{\pi(1-e^{-\gamma t})}\int d^2\beta'\,\exp\!\Big[-\frac{2|\boldsymbol{\beta}-e^{-\gamma t/2}\boldsymbol{\beta}'|^2}{1-e^{-\gamma t}}\Big]\,W_0(\boldsymbol{\beta}')\,, \tag{S21}$$

which contracts the state toward the origin with rate $\gamma/2$ while convolving it with a Gaussian of variance $(1-e^{-\gamma t})/4$ per quadrature. The damping part $e^{\gamma t}W_0(e^{\gamma t/2}\boldsymbol{\beta})$ preserves the integrated negativity $\delta_-(t) = \int |W_t|_- d^2\beta$ *exactly* (phase-space contraction compensated by amplitude renormalization), so only the diffusion erodes it. Expanding (S21) to first order in γt and verifying the result against the exact Lindblad flow in the truncated Fock space (Section S11) gives $\partial_t W|_{t=0} = \gamma\,[W_0 + \frac{1}{2}(\boldsymbol{\beta}\cdot\nabla)W_0 + \frac{1}{8}\Delta W_0]$. With $W_0 = 0$ on the boundary

$\partial\Omega$ of the negativity region $\Omega = \{W_0 < 0\}$, the divergence theorem gives $\int_\Omega W_0 = -\delta_-(0)$ and $\int_\Omega (\boldsymbol{\beta}\cdot\nabla) W_0 = 2\delta_-(0)$, and hence

$$\kappa_{\text{single}} \;=\; -\frac{1}{\gamma}\,\frac{d\ln\delta_-}{dt}\bigg|_{t=0} \;=\; \frac{1}{8}\,\frac{\int_\Omega \Delta W_0\, d^2\beta}{\delta_-(0)} \;=\; \frac{1}{8}\,\frac{\oint_{\partial\Omega}(\partial W_0/\partial n)\, ds}{\delta_-(0)}\,, \tag{S22}$$

i.e. the negativity decay rate under photon loss is set by the flux of the Laplacian through the negativity region, normalized by the negativity itself. For the truncated even cat (truncated-displacement construction, $N_{\max} = 2$, $\alpha = 1$; amplitudes $c_0 = 0.7462$, $c_2 = 0.6658$ on $|0\rangle, |2\rangle$) the Wigner function is $W_0 = (2/\pi)e^{-2r} g$ with $g = 1 - 8c_2^2\, r + 8c_2^2\, r^2 + 4\sqrt{2}\, c_0 c_2\, r\cos 2\varphi$ ($r = |\boldsymbol{\beta}|^2$), the negativity boundary is the explicit curve $8c_2^2 r^2 + (4\sqrt{2}\, c_0 c_2 \cos 2\varphi - 8c_2^2) r + 1 = 0$, and (S22) evaluates to $\kappa_{\text{single}} = 3.411$ — matching the numerically exact short-time slope (3.411); the earlier estimate 3.34 was limited by the 31×31 Wigner grid. In the large-cat limit the same identity returns the textbook value: the numerically exact dynamics give $\kappa = 8.54$ ($\alpha = 2$) and $\kappa = 17.97$ ($\alpha = 3$), converging to $2|\alpha|^2$; the asymptotic formula $2\gamma|\alpha|^2/\gamma = 2.0$ underestimates the truncated state by $\approx 1.71\times$ and is not the correct reference here. The single-particle analogue, $\Lambda_1 = \frac{1}{T}\int_0^T dt\, \sum_{j\neq 0}\gamma_j |\langle j|U(t)|0\rangle|^2$, provides a strict lower bound: $\Lambda_1 = 0.013$ (2 cells) and 0.052 (3–4 cells) lie below the measured residuals at every size. An independent recomputation of (S20) with the *production* parameters (Kerr term $k_{\text{Kerr}} = 0.5$ on the edge site, edge-low-loss rates, zero disorder) gives $\Lambda_2^{\text{coh}} = 0.012, 0.095, 0.095$ (two, three, four cells; Table S16). The structural content that survives independently is (i) the strict lower bound $\Lambda_1 < \gamma_{\text{eb}}^{\text{meas}}$ at every size; (ii) the *exact* three/four-cell degeneracy, $\Lambda_2^{\text{coh}}(3\text{c}) = \Lambda_2^{\text{coh}}(4\text{c})$, mirroring $\gamma_{\text{eb}}^{\text{meas}}(3\text{c}) \approx \gamma_{\text{eb}}^{\text{meas}}(4\text{c})$; and (iii) the boundary jump between two and three cells, present in Λ_2^{coh} ($0.012 \to 0.095$) but overpredicted in magnitude (a factor 7.9 vs the measured 2.79). The prefactor $\kappa = \gamma_{\text{eb}}^{\text{meas}}/\Lambda_2^{\text{coh}}$ is *not* size-universal: $\kappa = 11.8, 4.2, 4.2$ across sizes, a factor 2.8 between two and three cells. The closed form (S20) is therefore a *per-size calibration*, not a universal prediction: γ_{eb} is reproduced at each size only because κ is defined as the ratio, and no single κ transfers across sizes. The independent calculation computes the two-photon dwell under one explicit convention. It reproduces the strict single-particle lower bound and the exact three/four-cell degeneracy, but the inferred coefficient is size-dependent; the relation therefore provides a per-size calibration rather than a universal prediction of the finite-time decay. The exact single-site reference (S22) ($\kappa_{\text{single}} = 3.411$) remains a closed-form result for the bare-site cat, whereas the chain dynamics contain additional finite-time and many-body contributions.

TABLE S16. Independent recomputation of the edge–bulk rate (S20) at zero disorder (edge-low-loss profile): Λ_1 is the strict single-particle lower bound, Λ_2^{coh} the rate-weighted two-photon dwell under the production parameters (Kerr $k_{\mathrm{Kerr}} = 0.5$; $\sum_{i\le j}$ convention), $\Lambda_2^{\mathrm{coh,lin}}$ the same integral with the Kerr term omitted (linear two-step chain), $\gamma_{\mathrm{eb}}^{\mathrm{meas}} = \Gamma_{\mathrm{meas}} - \Gamma_{\mathrm{eff}}$ the measured residual, and $\kappa = \gamma_{\mathrm{eb}}^{\mathrm{meas}}/\Lambda_2^{\mathrm{coh}}$ the per-size prefactor. The exact three/four-cell degeneracy is reproduced under every convention; the prefactor is size-dependent, so the closed form is a per-size calibration rather than a universal prediction (see text).

Cells	Λ_1	Λ_2^{coh}	$\Lambda_2^{\mathrm{coh,lin}}$	$\gamma_{\mathrm{eb}}^{\mathrm{meas}}$	κ	$\kappa\Lambda_2^{\mathrm{coh}}/\gamma_{\mathrm{eb}}^{\mathrm{meas}}$
2	0.0128	0.0120	0.0256	0.1419	11.8	1.00
3	0.0523	0.0950	0.1046	0.3962	4.2	1.00
4	0.0523	0.0950	0.1046	0.3962	4.2	1.00

The independent calculation is therefore used only as a structural diagnostic. It supports the strict lower bound $\Lambda_1 < \gamma_{\mathrm{eb}}^{\mathrm{meas}}$ and the three/four-cell degeneracy, but it does not predict the measured finite-time decay without a size-specific coefficient. We do not report a per-realization table and do not interpret the ratio as a parameter-free law. The residual contains additional many-body and finite-time effects, including Gaussianization of the local marginal.

c. Sector dependence of the prefactor. The closed form (S20) with the per-size prefactor $\kappa_N = 11.8, 4.2, 4.2$ is calibrated in the topological sector (the sector of the main-text contrast). Applying the *same* formula with the same per-size κ_N to the trivial sector (control; A and B interchanged) shows the limits of the linear-loss model. With the trivial-sector parameters the site-0 mode is far more edge-concentrated ($|w_0|^2 = 0.30, 0.25, 0.17$ at $2, 3, 4$ cells vs. 0.013 topological) and the measured decay is slower than the transferred prediction by factors 3.7, 1.45, and 1.51, i.e. an effective prefactor $\kappa_{\mathrm{eff}} = 2.86, 2.69, 2.48$ that decreases with size. The trivial sector lies below the exact bare-site reference $\kappa_{\mathrm{single}} = 3.411$ (0.84, 0.79, 0.73×): the finite-time drive returns part of the leaked population within the period, so the trivial mode does not convert its loss exposure as efficiently as the continuously-exposed bare site, whereas the topological per-size κ_N exceeds κ_{single} (3.46, 1.22, 1.22×) — the excess measures the many-body and finite-time contributions that (S20) absorbs into the calibration. The mode-structure dependence is the *ratio* $\kappa_{\mathrm{topo}}/\kappa_{\mathrm{eff}} = 4.1, 1.55, 1.68$: the delocalized topo-

logical edge mode ($|w_0|^2 = 0.013$) converts its small two-photon dwell ($\Lambda_2^{\rm coh} = 0.0120$) into coherence destruction more efficiently than the trivial mode, which spreads the two-photon state over a 13× (2 cells) or 1.7× (3–4 cells) larger dwell ($\Lambda_2^{\rm coh} = 0.158, 0.161, 0.161$), diluting the coherence over many pair configurations and lowering the per-dwell efficiency. The closed form reproduces the measured *ordering* of the two-cell rates (calibrated topological $0.337 <$ transferred trivial 1.948; measured $0.337 < 0.532$) but *not* the three- and four-cell sign change (calibrated topological $0.591 <$ transferred trivial 0.772 and 0.805; measured $0.591 > 0.533$): the sign change is a many-body Gaussianization effect that the linear-loss dwell cannot produce, and it is exactly the regime in which the second consequence of the lemma (delocalization → Gaussianization) dominates over the first.

The two-to-three-cell jump is consistent with a boundary effect of the mode delocalization. The inter-cell Floquet step transfers the edge-site excitation to site $j = 3$ with a period-averaged dwell $\langle P_3 \rangle_T = 0.198$; the single-particle dwell profile $[0.739, 0.064, 0.000, 0.198, \dots]$ is identical at all chain lengths (the first cell and its inter-cell partner fully determine the short-time propagation). In the two-cell chain site 3 is the opposite edge ($\gamma_3 = 10^{-4}$), so the visiting population is shielded; from three cells on it is a bulk site ($\gamma_3 = 0.2$) and the same dwell is exposed to the bulk loss rate. The rate-weighted integrals Λ_1 and $\Lambda_2^{\rm coh}$ inherit exactly this boundary dependence — they jump once at three cells and stay constant thereafter — which matches the observed pattern of $\gamma_{\rm eb}$. The delocalization of the edge mode into the bulk is thus not size-dependent through the mode weights (which are pinned by the local boundary), but through the *identity* of the sites the propagating excitation visits: longer chains place the site reached by the inter-cell step deeper in the lossy bulk.

This lemma is the negative side of the mixed-state-SPT lesson discussed in the main text: the localizer gap pins the coherent sector, but the negativity survives only to the extent that its generating mode stays localized at the measured site; the boundary condition of the main text—that coherent topology and open-system resource preservation are logically distinct—is thereby given a quantitative, falsifiable form, consistent with the validated collective-mode null result (Section S9 P): the site-basis observable fails to commute with the delocalized mode structure, so its protection cannot be inferred from the single-particle invariant. The null result is in fact consistent with Eq. (S19): the collective edge mode is the same sector vector c with bulk participation $1 - |c_0|^2 = 0.987$, so its marginal is already Gaussian at $t = 0$ ($\delta_-(0) = 0.000\,000$ on the identical cat-state initial condition, vs. $\delta_-(0) = 0.1962$ for the

TABLE S17. Collective-mode Wigner null and dwell diagnostic (edge-low-loss, zero disorder; collective edge mode c is the same sector vector used for the local edge-mode construction). The collective marginal is Gaussian at $t = 0$ ($\delta_-(0) = 0.000\,000$) and remains zero in all tested records. The reported dwell values are structural diagnostics; no universal prefactor or inferred $\gamma_{\rm eb}$ is claimed.

Quantity	Collective edge mode	Local site 0
Edge weight $\lvert c_0\rvert^2 = \lvert u_{e,0}\rvert^2$	0.013 43	0.013 43
Bulk participation $1 - \lvert c_0\rvert^2$	0.9866	0.9866
$\delta_-(0)$ (identical cat)	0.000 000	0.1962
Λ_1	0.195	0.013 (2c) / 0.052 (3–4c)
$\Lambda_2^{\rm coh}$	0.3925 (2c) / 0.3952 (3–4c)	0.012 (2c) / 0.095 (3–4c)
$\gamma_{\rm eb}$ inference	n/a ($\delta_-(0) = 0$)	size-specific
$\Gamma_{\rm eff}$ diagnostic	not inferred	0.337 (2c) / 0.591 (3–4c)
$\delta_-(T)$ (data)	0.000 000 in 320/320	0.0237 (2c) / 0.0048 (3c)

local site), and the bound is satisfied trivially, $\delta_-(T) = 0 \le \delta_-(0)\, e^{-\Gamma_{\rm eff} T} = 0$, in every record (80/80), at both Fock truncations $N_{\max} = 2, 3$ and both sizes. The closed form Eq. (S20) over-determines this null result: the collective two-photon state $|c, c\rangle$ is itself an (almost) stationary eigenmode of the two-particle Floquet operator, so its dwell is static and fully bulk-exposed, $\Lambda_2^{\rm coh} = 0.3925$ for two cells and 0.3952 for three–four cells (with $\Lambda_1 = 0.195$); no $\gamma_{\rm eb}$ or total decay rate is inferred from the collective null $\Gamma_{\rm eff}^{\rm tot} \approx 1.32$ — the largest in the model, $\approx 2.2\times$ the local three-cell rate and $\approx 6.8\times$ the static mode-weighted floor. Even had the collective marginal possessed initial negativity, the bound would therefore decay it at the fastest available rate; as it is, zero initial resource ($\delta_-(0) = 0$) makes the inequality $\delta_-(T) = 0 \le 0\, e^{-\Gamma_{\rm eff}^{\rm tot} T} = 0$ satisfied in every record. The local mode retains a positive weight $|u_{e,0}|^2 = 0.013 > 0$ and is the only marginal able to sustain the observed positive contrast under edge-selective loss. The full numeric record is compiled in Table S17.

S10. PHYSICAL PARAMETER MAPPING AND HARDWARE BOUNDS

The main text uses normalized model coordinates ($T = 2\pi$, all rates in units of the drive angular frequency $\omega_d = 2\pi/T$). This section gives an *illustrative* mapping to concrete

hardware magnitudes so that the dimensionless content of the study can be compared with real platforms. It is not a calibration: no optimal operating point is asserted, and the values below are representative order-of-magnitude examples chosen to preserve the dimensionless ratios of Table S1.

Two platforms preserve the same dimensionless ratios: a superconducting circuit array and an optomechanical array. The mapping is fixed by the drive frequency $\omega_d = 2\pi/T$; any normalized rate Γ becomes $\Gamma\,\omega_d$. For the superconducting platform we take $\omega_d/2\pi = 100\,\mathrm{MHz}$ ($T = 10\,\mathrm{ns}$), giving hopping $J_B = 1.2\,\omega_d \simeq 754\,\mathrm{MHz}$, Kerr $\chi = 0.5\,\omega_d \simeq 314\,\mathrm{MHz}$, bulk loss $\gamma_{\mathrm{bulk}} = 0.2\,\omega_d \simeq 126\,\mathrm{MHz}$, and edge loss $\gamma_{\mathrm{edge}} = 10^{-4}\,\omega_d \simeq 63\,\mathrm{kHz}$. The intracell hopping is $J_A = 0.1\,\omega_d \simeq 63\,\mathrm{MHz}$. For the optomechanical platform we take $\omega_d/2\pi = 10\,\mathrm{MHz}$ ($T = 100\,\mathrm{ns}$), giving $J_B \simeq 75\,\mathrm{MHz}$, $\chi \simeq 31\,\mathrm{MHz}$, $\gamma_{\mathrm{bulk}} \simeq 13\,\mathrm{MHz}$, $\gamma_{\mathrm{edge}} \simeq 6.3\,\mathrm{kHz}$, and $J_A \simeq 6.3\,\mathrm{MHz}$. The dimensionless ratios preserved in both rows are $\chi/J_B = 0.42$, $\gamma_{\mathrm{bulk}}/\chi = 0.4$, and $\gamma_{\mathrm{edge}}/\gamma_{\mathrm{bulk}} = 5 \times 10^{-4}$.

The mapped scales are consistent with known hardware bounds. In superconducting circuits, hopping amplitudes of tens to hundreds of megahertz and Kerr self-energies of a few hundred megahertz are within reach of tunable-coupler and transmon-based arrays [14]; a bulk loss of order a hundred megahertz requires deliberately engineered dissipation (e.g., a Purcell-filtered resistor), while an edge loss in the tens of kilohertz corresponds to a quality factor $Q_{\mathrm{edge}} = \omega_d/\gamma_{\mathrm{edge}} \sim 10^4$ at the working frequency, which is routine for a terminated but uncoupled site. In optomechanical arrays [15], phonon hopping in the megahertz range and engineered mechanical loss in the kilohertz–megahertz range are standard; the Kerr strength $\chi \sim 31\,\mathrm{MHz}$ corresponds to a strong but reachable mechanical nonlinearity or to an auxiliary nonlinear element. The two-step Floquet drive is realized by periodically switching the effective hopping amplitudes (e.g., by modulating coupler fluxes or a parametric drive), a standard technique in both platforms.

A concrete measurement chain connects the dimensionless observables to these platforms. The site Wigner function is a direct-measurement target: the parity-based reconstruction of Lutterbach and David [16] maps the Wigner function onto measurable displaced parities of a single mode, and circuit QED implements the same displacement-and-readout primitive, where Schrödinger cat states as large as 111 photons have already been prepared and controlled [17]. The grid-free witness of Section S9 N is a fixed functional of displaced photon-number statistics $\langle n|D^\dagger(\alpha)\rho D(\alpha)|n\rangle$, so certifying the two-cell signal requires only a

finite set of displaced measurements rather than full tomography; site selectivity in the lattice then follows from the frequency-resolved cavity modes addressed by a shared ancilla, as in standard multiresonator circuit-QED readout [14]. The engineered bulk loss of Table S1 corresponds to dissipative stabilization of a dynamical manifold—a reservoir-engineering protocol already demonstrated in circuit QED [18]. On the optomechanical side, the equivalent mechanical-quadrature reconstruction belongs to standard cavity-optomechanical readout [19]. These primitives set the measurement overhead a concrete device would carry; they do not alter the back-action and heating exclusions stated below.

These primitives turn the reported effects into concrete testable predictions, summarized in Table S18. The most accessible is the two-cell witness asymmetry: certifying the topological-sector signal requires only displaced photon-number statistics at a finite set of displacements, with expected single-shot certification rates of 82.4 versus 10.8 percent between sectors (Section S9 N)—a few hundred repetitions separate the sectors with no tomography and no phase-space grid. The full contrast sign pattern (positive at two cells, negative at three to five cells) is a tomography-level prediction requiring sub-10^{-3} resolution at the larger sizes, where the solver-level branch sits at -9×10^{-5} to -6×10^{-4}; the Kerr-free stratum predicts the same sign pattern with the exact two-cell contrast reduced a factor 2.1 (Section S9 D). Every entry is conditional on stroboscopic readout at $t = T$ and on the structured loss profile ($\gamma_{\mathrm{edge}}/\gamma_{\mathrm{bulk}} = 5 \times 10^{-4}$): the micromotion analysis predicts sign changes within the period under every tested condition, and the loss-ratio scan predicts the signal to vanish once the ratio reaches 0.5 (Section S9 I).

Two bounds follow. First, the reported contrasts are ratios of single-period observables, so they are invariant under a uniform rescaling of ω_d; the physical time over which the sign flip occurs is $T \sim 10\,\mathrm{ns}$ to $100\,\mathrm{ns}$ in these examples, well below typical coherence times ($T_1, T_2 \gtrsim 1\,\mu\mathrm{s}$ for circuits, $T_2 \gtrsim 10\,\mu\mathrm{s}$ for optomechanics). Second, the normalized study deliberately does not include the measurement back-action or drive-induced heating that a concrete device would add; the mapping therefore bounds what the coherent model predicts in an idealized device, not a hardware result. The table row "Physical calibration and optimality" in Table S19 reflects this boundary: the mapping is illustrative and the parameters are not optimized.

TABLE S18. Testable predictions of the present study. All values requote production results; none is a new fit.

Prediction	Measurement	Expected outcome
Two-cell witness asymmetry	Displaced photon counting, no tomography	82.4% vs 10.8% certification (disjoint Wilson intervals)
Contrast sign pattern, 2–5 cells	Site-resolved Wigner tomography at $t = T$	$+0.017$ at two cells; -0.002 at three to five cells
Kerr-free sign pattern	Same, with nonlinearity off	Same signs; two-cell exact contrast 2.1× smaller
Uniform-loss null	Same, uniform profile	Contrast at 10^{-6} level at $t = T$
Stroboscopic conditioning	Readout at $t \neq T$	Sign changes within every period
Loss-ratio dependence	Vary edge-to-bulk ratio	Signal gone by ratio 0.5
Five-cell sign	Ten-mode tomography	Negative, inside frozen interval

A. Step-by-step measurement protocol

The following protocol executes the tests of Table S18 on either platform row above, in normalized units throughout. It is prescriptive: each step states its setting, its check, and its acceptance criterion. No step requires fitting a model to the data.

1. *Prerequisites.* An array of 4–10 bosonic modes with site-resolved readout, tunable nearest-neighbour couplings, a Kerr nonlinearity addressable on the edge site, and engineered loss with an edge-to-bulk contrast of at least two orders of magnitude (production uses $\gamma_{\text{edge}}/\gamma_{\text{bulk}} = 5 \times 10^{-4}$; the signal is gone by 0.5). Multimode cats have been stabilized by normal-mode dissipation engineering [20] and cat manifolds by nonlinear reservoir engineering in trapped ions and superconducting circuits [21], so the cat-plus-engineered-loss combination is an established primitive rather than a speculative one.

2. *Coherent calibration.* Set the hoppings to the working points $(J_A, J_B) = (0.1, 1.2)$ (chiral) and $(1.2, 0.1)$ (control). Accept when spectroscopy reproduces the minimum zero and π quasienergy gaps 2.1991 and 0.3142 and the open chain shows edge modes at the chiral point only.

3. *Loss engineering.* Set $\gamma_{\text{bulk}} = 0.2$ on bulk sites and $\gamma_{\text{edge}} = 10^{-4}$ on boundary sites (drive units). Accept when ringdown measurements reproduce both rates within the tolerance that keeps their ratio below 10^{-2}.

4. *State preparation.* Prepare the even cat of amplitude $\alpha = 1$ on site 0. Accept when the pre-drive local Wigner negativity reads $\delta_-(0) = 0.1962$ within tomography error: this verifies the resource before the drive acts.

5. *Single-period drive.* Run exactly one two-step square period T and read out locked to the drive phase at $t = T$. This lock is mandatory, not advisory: the micromotion analysis predicts sign changes within the period under every tested condition.

6. *Witness measurement (near-term target).* Perform displaced photon counting and evaluate the F_1 witness against $\omega_1 = 0.5$. Separating the expected rates (82.4% vs 10.8%) needs 7 shots per sector at 80% power and 10 at 95% (two-sided $\alpha = 0.05$); 300 shots give Wilson half-widths below 0.05 on both rates. Accept when the two Wilson intervals are disjoint and ordered as predicted (topological rate above trivial).

7. *Tomography measurement (full pattern).* Reconstruct the edge-site Wigner function on a 31×31 grid (961 settings per function) for both sectors and form the contrast. Accept when the measured contrast reads $+0.0167 \pm 0.0011$ (two cells), -0.0021 ± 0.0011 (three and four cells), -0.0022 ± 0.0007 (five cells); accept ≈ 0 under the uniform profile at $t = T$.

8. *Disorder and sweep repetition.* Repeat steps 4–7 over disorder amplitudes 0–1.5 (tunable couplers or device ensemble, 500 matched units per setting at two and three cells as in production) and along $J_A + J_B = 1.3$ for the transition test. Accept when the contrast stays continuous through all three gap closings with a zero window straddling the ν_0 transition.

9. *Falsification criteria and non-goals.* The boundary claim fails if any of the following is observed: contrast tracking the invariant across the transition, nonzero uniform-loss contrast at $t = T$, or a five-cell sign outside the frozen interval. Not required: steady-state measurements (single period only), back-action or heating characterization beyond reporting them as systematics (not modeled here), or device optimization (parameters are illustrative).

S11. SHORT-TIME LINDBLAD EXPANSION IN THE TRUNCATED SPACE

The short-time expansion underlying (S22) can be verified in closed form against the exact Lindblad flow. In the truncated $N_{\max} = 2$ space the even cat is $|\psi\rangle = c_0|0\rangle + c_2|2\rangle$ with $c_0 = 0.7462$, $c_2 = 0.6658$ (the truncated-displacement construction, $\alpha = 1$). For comparison, the ideal untruncated even cat at this amplitude has Fock weights 64.8 % ($n = 0$), 32.4 % ($n = 2$), and 2.79 % ($n \geq 4$), so the production construction excludes the $n \geq 4$ tail by design. The amplitude-damping channel $\dot{\rho} = \gamma(a\rho a^\dagger - \frac{1}{2}\{a^\dagger a, \rho\})$ closes on the diagonal and coherence elements

$$\rho_{22}(t) = c_2^2 e^{-2\gamma t}, \qquad \rho_{11}(t) = 2c_2^2 \gamma t\, e^{-\gamma t}, \qquad \rho_{02}(t) = e^{-\gamma t}\big[c_0 c_2 + c_2^2 \gamma^2 t^2\big], \tag{S23}$$

with $\rho_{00} = 1 - \rho_{11} - \rho_{22}$. Inserting these into the Wigner function $W_t = (2/\pi)e^{-2r}[\rho_{00} + \rho_{11}(4r - 1) + \rho_{22}(1 - 8r + 8r^2) + 4\sqrt{2}\,\rho_{02}(x^2 - y^2)]$ and expanding to first order in γt gives, identically in r and $x^2 - y^2$, $\partial_t W|_{t=0} = \gamma[W_0 + \frac{1}{2}(\boldsymbol{\beta} \cdot \nabla)W_0 + \frac{1}{8}\Delta W_0]$ — the same expansion obtained from (S21). The two independent derivations (Gaussian-channel solution and exact Lindblad flow) therefore agree to all orders needed for (S22).

S12. SCOPE AND LIMITATIONS

The limits in Table S19 confine the conclusions to the coherent-sector calculations, the finite-time open-system comparison and the specified deterministic calculations.

[1] J. K. Asbóth, B. Tarasinski, and P. Delplace, Physical Review B **90**, 125143 (2014).

[2] J. Dalibard, Y. Castin, and K. Mølmer, Physical Review Letters **68**, 580 (1992).

TABLE S19. Scope of the conclusions for the results presented in the main text and Supplemental Material.

Quantity	**Interpretation**
Chiral bulk invariant	Reproduced by two independent implementations (frame winding and spectral localizer; Eq. (S16)).
Finite-chain edge state	Used as a model-level observable.
Paired trajectories and finite-value analysis	Supported for the stated comparison.
Size-robust local Wigner protection	Not supported by the data.
Passive edge-mode reference	Not established.
Cutoff convergence of reference	Not established.
Local-negativity cutoff dependence	Reported descriptively; not converged.
Wigner-grid discretization of the integral	Within $\pm 3\%$ of the finest grid.
Trajectory-count sensitivity (nested)	$\approx 4\%$ on a fixed realization; contrast-level ladder in Section S9 B.
Collective-mode Wigner negativity	Not established; measured as null for both preparations (common-site and mode-resolved) (Section S9 P).
Multi-period behavior	No sign inversion up to eight periods; two-cell magnitude revives at four periods (Section S9 H); not a monotone protection signal.
Edge-to-bulk loss-ratio dependence	Positive four-mode contrast requires the sharp engineered profile; lost for ratio 5×10^{-1} (Section S9 I).
Eight-mode statistical power	Resolves $\lvert\Delta\rvert \gtrsim 0.000\,77$ at $N_{\mathrm{MC}} = 2000$ ($n = 2000$, $\Delta_{\min}$ for power 0.80); observed mean $-0.002\,01 \approx 2.6\times$ the resolution ($1.35\times$ at the original $n = 500$; Section S9 F).
Correspondence with the bulk invariant	Not established: the contrast is continuous at all three gap closings of the dimerization sweep (resid-

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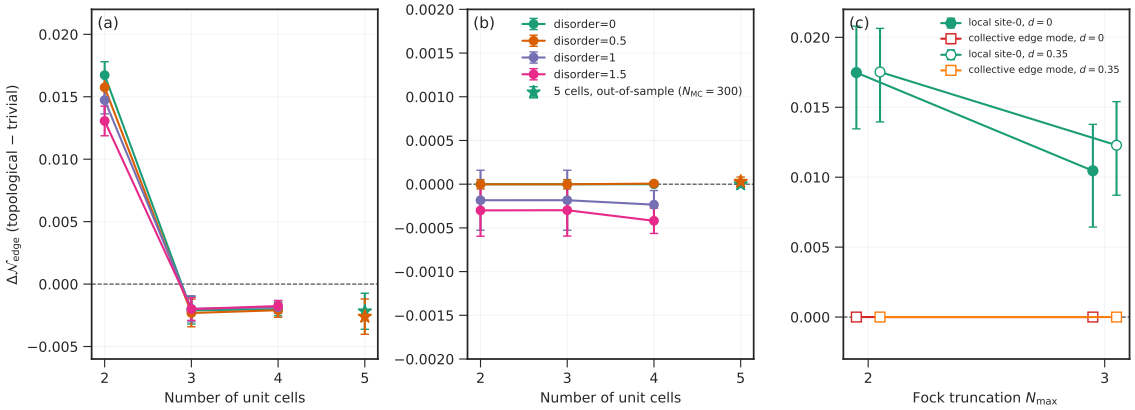


FIG. S4. Physical-driveness scan across the same Rev. Phys. conditions used in the revision-scan summary (loss profile, disorder, and engineered edge/bulk rates). The panel reproduces the production contrast estimate (Monte Carlo, $n_{\mathrm{traj}} = 10$) whose solver agreement is cross-checked in the main text; it is shown here as a figure-level record of the scan rather than as an additional result. Abbreviations and conventions follow Section S1.